

Onboarding and Orientation

- Workbook -

ENGINEERING FOR SUSTAINABILITY



RHINE-WAAL UNIVERSITY OF APPLIED SCIENCES



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1 Sustainability & Common Goals

1 Disclaimer

These notes are exclusively dedicated to usage within the framework of the lectures at the Rhine-Waal University of Applied Sciences. An unauthorised disclosure is not permitted. If you obtained these notes without taking part in the named course or without the explicit permission of the author, please write an e-mail to Sekretariat-TuB@hochschule-rhein-waal.de

2 Welcome on Campus

Welcome to the study programme **Engineering for Sustainability**! A study programme where your ideas and your personality is shaping the content - not the other way around! Probably you already have read the official information, e.g.

- Official Information on the homepage [1]
- Module Handbook defining the content [4]
- Exam regulations [2]

Also you know already about the web-portal for keeping track on the contents of the respective course in *moodle*. Surely, still a lot of questions are left unanswered at the moment, as the given study programme is no ordinary one. For answering some of those questions, the module 2203 - *Onboarding and Orientation* is designed. The document at hand should help you by structuring the information and also helping you to define your personal goals - as a handwritten notes are usually perceived more personal than typing on a computer (which we will also do a lot in the upcoming courses), it is advised to either print it out to fill in several informations using a pencil or use a digital pen on a tablet-pc - which ever you rate as the more sustainable solution (also being your first sustainability decision - so be prepared for an upcoming discussion). However, you had very distinctive reasons for enrolling in **Engineering for Sustainability**. Reasons that should be honoured and put in writing at the beginning of the study, to make sure that these reasons remain in the focus of your personal development over the next years. Also, those reasons should be communicated to your *personal mentor* later on for offering guidance on how to achieve what **you** are planning to do with your life, or as Niklas Luhmann¹ put it

'Without writing, one cannot think; at least not in a sophisticated, coherent manner.'

¹Original quote: „Ohne zu schreiben, kann man nicht denken; jedenfalls nicht in anspruchsvoller, anschlussfähiger Weise.“, see [42], p.222

2.1 Rhine-Waal University: Ordinary is Elsewhere

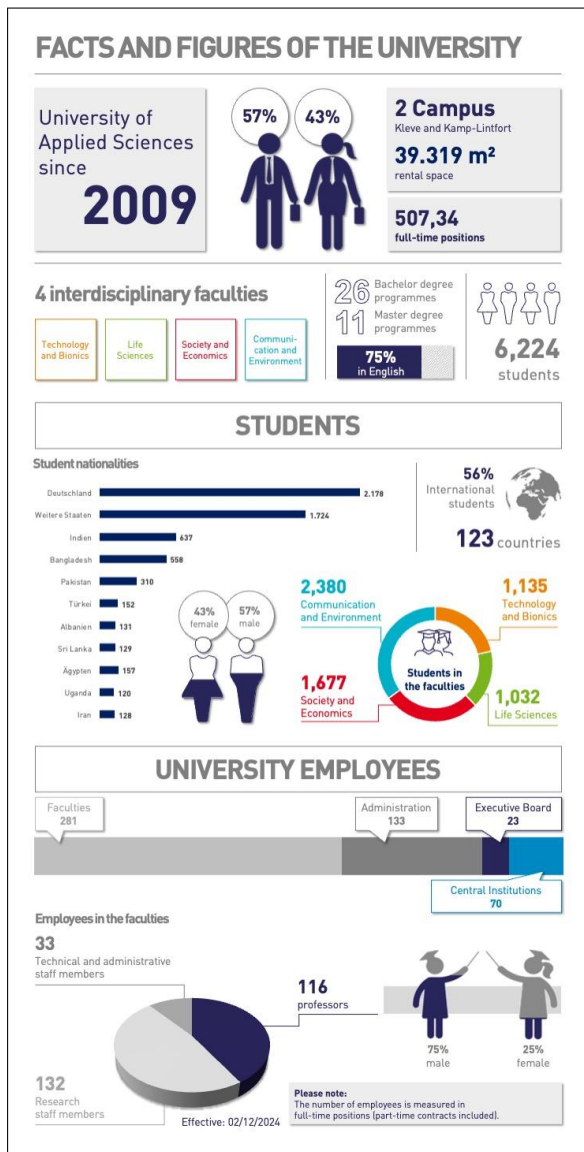


Figure 1.1: Infografik Hochschule in Zahlen
Stand 12/2024, from [14]

You arrived at the Rhine-Waal University and supposedly you already visited the campus, e.g. taking part in the activities of the freshers week <https://www.hochschule-rhein-waal.de/en/academics/students/welcome/freshers-week> or simply strolling along the Spoy-channel. What you will have noticed is what as a number can be taken from the middle right part of fig. 1.1 - the HSRW has a unique distribution of the nationalities where the students come from, or as we put it in the accreditation report [27], p.4f:

'Innovative, interdisciplinary, and international

Rhine-Waal University of Applied Sciences was founded in 2009 and stands for innovative, interdisciplinary, and international education for young people. At its locations in Kleve and Kamp-Lintfort, the university offers high-quality, scientifically and application-oriented, internationally focused study programs as well as practical teaching and research. The degree programs focus on the natural sciences and technology. In addition, the university offers degree programs in the social sciences and health sciences.

The interdisciplinary nature of the degree programs The interdisciplinary nature of the courses is also reflected in the names of the four faculties. The Kleve campus is home to the faculties of Life Sciences, Technology and Bionics, and Society and Economics. The Faculty of Communication and Environment is located on the Kamp-Lintfort campus. A total of 25 bachelor's and 11 master's degree programs are offered. In-

novative, forward-looking subject combinations in the curricula, interdisciplinarity of the fields of knowledge, and thinking outside the box are common features of the degree programs. Approximately 75 percent of the degree programs are taught in English, and more than 6,400 students from over 120 countries are now studying together in an environment that many have specifically chosen for its cultural diversity and internationality. This provides students with intercultural exchange and optimal preparation for the national and international job market.

The university expressly commits itself to cosmopolitanism and the values associated with it by participating in the national code of conduct for foreign students at German universities (since



2020), the Germany Scholarship (since 2016), and the nationwide campaign “Cosmopolitan Universities Against Xenophobia” (2020). The university also holds Fairtrade University certificates (2017, recertified in 2020), the first of its kind in North Rhine-Westphalia, for “Shaping Diversity” (2018, recertified in 2022) and as a family-friendly university (2015, recertified in 2018 and 2021) demonstrate and recognize the particularly active commitment of Rhine-Waal University of Applied Sciences to internationality, equality, and sustainability.’

Exercise 1:

Take a look at the original data of the HSRW as published in [13]. How many people from your home country were enrolled in one of the universities study course? At which faculty are the most and at which are the least students from your country (why)?

The immense international character and hereby tremendous diversity deserves special consideration, as this is a very valuable resource, in particular when dealing with the multidisciplinary of scientific projects. In process gains are increased creativity and, under some boundary conditions, increased satisfaction in particular for more complex problems, communication and coordination need an increased amount of effort, though [62] [63]. The surrounding culture of ones primary (and secondary) socialisation processes has an impact on cognition, emotion, and motivation [48] [44], or by as what is known as Hebb’s mechanisms *Cells that fire together, wire together*, the specific usage of our brain has physical consequences (which takes time to change), even the language has a significant and measurable impact [64] - that is why understanding a culture usually starts with learning the language. Also, diversity has to be treated with care, as there is even a possibility of lowering perceived team effectiveness if some prerequisites are not fulfilled, e.g. for a beneficial cooperation in diverse scientific groups, beside a maintaining focus on the common goal, mutual respect and trust between members is fundamental [61]. Newer findings have shown that friendship is not only ‘correlated with a sense of well-being across the life span’ [34],(p.355) but can also be predicted by neural similarity [59], when also taking the plasticity of the human brain into account [43] [32], the author states the (potentially questionable) combined hypotheses²:

Friendship scales with increased mutual understanding and mutual understanding is promoted by constructive communication. As peace between friends is a natural attribute, in consequence good communication leads to peace ;-).

Peace is (besides justice and strengthening institutions)³ included in the united nations sustainable development goals - as *engineers for sustainability* - this is our obligation.

As this issue is so important for our cooperation for our common sustainable goals, within the study course, it is specifically addressed in a separate course in the first semester in *Modul 2201 Fundamentals of Personal and Social Competences*. So, without going into too much detail on communication theories, be invited to experiment on communication and treat its natural shortcomings as a feature, or as quoted in [42], p.222f:

²Certainly not fulfilling the criteria of a good hypothesis of the scientific branch of peace studies, see [8]

³Original quotation of goal 16 according to [46]: ‘*Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels*’

Communication is one of the basic requirements for partners to be able to surprise each other. Only in this way is it possible to generate information in the other person. (...) It is not necessary for both partners to use the same comparison scheme; the surprise effect actually increases when this is not the case and it is considered a coincidence that a message means something or is even useful against a backdrop of other possibilities.

Also be aware that, rooting in different personal and cultural backgrounds of the communicating persons, occasional misunderstanding is inevitable. So carrying *Hanlon's razor* [7] in your pocket as a quick tool for protecting your self-esteem and your personal view of the world might be suitable:

'Never attribute to malice that which is adequately explained by stupidity.'

Also, nobody is safe from finding her-, or himself on both sides of the named razor.

Exercise 2:

Let's put what we have learned into your personal context, and look at your cultural background and what you might be confronted with for a quick cut with Hanlon's razor. Below, fig. 1.2 shows some random quotes on typical rules of conduct in businesses.

Enterprises in your home country: What are the 'dos & don'ts'? What is true, what is not?

India: If your business partner offers you a meal, never say "thank you" at the end of it, as this is considered a form of payment on your part and therefore insulting.

Japan: Japan is especially sensitive to the word "no." It is customary to respond with "yes" even if you disagree with what is being said

Italy: Punctuality is not a priority for Italians. Be patient and prepared for some delay when you start working with a new Italian partner.

Germany: When entering business meetings, allow the oldest person to enter the room first.

China: Chinese people appreciate a conservative dress code, and you should avoid physical contact while conversing.

Brazil: Business meetings often last longer than planned, but do not leave before the meeting has officially ended. Leaving early is considered rude.

France: It is vital to make appointments for both business and social occasions. It is not acceptable in France to drop in on someone unannounced.

Figure 1.2: Assumed business rules of conduct in different countries [9], own arrangement in given figure

Check the original source

[https://www.business.com/articles/
so-international-business-etiquette-from-around-the-world](https://www.business.com/articles/so-international-business-etiquette-from-around-the-world)

for quotes about the country you come from. What percentage of the assumed rules do you think is correct? What do you assume is the most misunderstood prejudice about your country? What is the most important thing to know about when others deal with members of your cultural group (which ever cultural group you identify with the most)?



2.2 Faculty of Technology and Bionics

Restructuring the degree programmes

The⁴ Faculty of Technology and Bionics, is one of the four faculties at Rhine-Waal University of Applied Sciences. The faculty has an interdisciplinary and international focus, which is reflected in the composition of its staff, the degree programme profiles, the dedication of its professors and the student body. Modern engineering sciences, bio-inspired technical solutions and products, sustainable Modern engineering sciences, bio-inspired technical solutions and products, sustainable and contemporary developments in materials, and the latest issues in science communication – these and many other topics are currently taught and offered in five English-language Bachelor's programmes and two English-language Master's programmes at the Faculty of Technology and Bionics:

- Biomaterials Science B.Sc
- Electrical and Electronics Engineering B.Sc.
- Industrial Engineering B.Sc.
- Mechanical Engineering B.Sc.
- Mechatronic Systems Engineering B.Sc.
- Bionics M.Sc.
- Mechanical Engineering M.Sc.⁵

(...) Various studies show that student numbers and the requirements profile expected of future graduates by industry will change, while at the same time there is a threat of a shortage of skilled workers on the labour market, especially in technical professions. This has prompted the faculty to consider reforming its bachelor's degree programmes in engineering from 2022 onwards. (...) In future, the Bachelor's degree programme profile will consist of the following three programmes:

- Nature Inspired Materials, B.Sc.: The programme covers biocompatible, biomimetic and nature-based materials and was already accredited when the faculty was founded under the name Biomaterials Science B.Sc.
- Engineering, B.Sc.: The programme combines various disciplines and forms a concept in which students have a high degree of freedom of choice and can select different specialisations after the orientation semester.
- Engineering for Sustainability, B.Sc.: This degree programme is an engineering programme focused on sustainability. 'Sustainability' represents both the orientation and the attitude of the teaching staff and students. The pedagogical and didactic approach therefore consistently focuses on project-based learning.

⁴whole subsection taken from [27] and translated to english

⁵The mentioned bachelor programmes are discontinued and phased out until 28.02.2030 and replaced by the programmes shown in fig. 1.3, heavily driven by a paradigm shift in university education, marked e.g. by the 'Recommendations for a sustainable approach to studying and teaching' by the german science council [72] and altered requirements of the society and industry on future engineers.

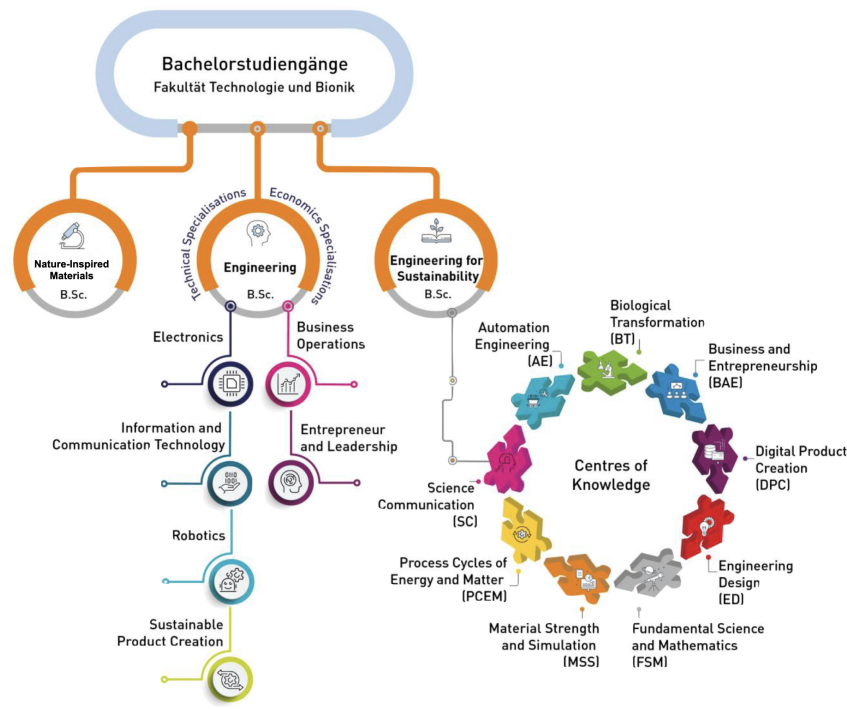


Figure 1.3: Overview on Bachelor degrees to be earned at the faculty technology and bionics (TuB) of the HSRW, taken from [27]

A Sustainable Approach to Studying and Teaching

The mentioned bachelor programmes are discontinued and phased out until 28.02.2030 and replaced by the programmes shown in fig. 1.3, heavily driven by a paradigm shift in university education, marked e.g. by the 'Recommendations for a sustainable approach to studying and teaching' by the German Science Council [72] and altered requirements of the society and industry on future engineers. Although the complete content of the cited recommendations can be seen as a guideline for the development of the study course *engineering for sustainability*, due to their relevance two quotes are mentioned here on behalf of the underlying spirit:

- 'Responsible personalities develop (...) less through guidelines and more through decision-making freedom and guidance.' (p. 31)
- 'Curricular offerings should (...) be more strongly oriented towards professional and academic socialisation.' (p. 37)

That summarises into one of the major messages for you as a student: You are an active part of the degree programme. It is wanted, expected, demanded and essentially needed for the success of the given course that you are a fully responsible member of the academic society. Although skills and competencies needed will develop over time, accepting that role can be done right from the start.



General Concept of the Study Programme Engineering for Sustainability

Brief profile of the Engineering for Sustainability degree programme⁶:

Key Features of the Study Programme Engineering for Sustainability

- **New approaches: Project-based teaching and learning in the Engineering for Sustainability degree programme**

The newly developed Engineering for Sustainability degree programme is a seven-semester, project-based engineering bachelor's degree programme with a professional, scientific focus. The focus on sustainability, already conveyed in the title, gives the programme a socially relevant meaning and points to its objective of placing technology-oriented solutions in a holistic, systemic and interdisciplinary context and evaluating them against this background. The project orientation as an essential didactic feature is taken up as a characteristic in the description of the degree programme, but is not addressed in the title in order to emphasise the focus on the specific objective.

- **University profile: Co-sponsor of the sustainability transformation at Rhine-Waal University of Applied Sciences**

The new degree programme significantly supports the sustainability transformation of Rhine-Waal University of Applied Sciences, as manifested in the TransRegINT project, the EUZENT institution and the 2030 University Development Plan (HEP 2030), and complements it with the necessary component of an engineering perspective. The inherent project character in the sense of the 'One Local anchoring is ensured by the inherent project character in the sense of "One Mission", the combined engineering education involving research projects and laboratories (e.g. FabLab), and development projects with partners from industry and public institutions based in the region. The local connection is intended to appeal not only to the declared international orientation but also to prospective students who feel connected to the region in order to achieve a balanced heterogeneity. The programme is intended to be offered in (...) English. In addition to its declared international orientation, the local connection is also intended to appeal to prospective students who feel connected to the region in order to achieve a balanced heterogeneity.

- **Faculty profile: Integration and shaping**

The new degree programme, in combination with the faculty's engineering degree programme, is a comprehensive concept designed to fully cater to the individual inclinations of potential students. The customisable freedoms of competence orientation for developing new, future problem-solving strategies in project-based engineering education do not require the comprehensive content of traditional degree programmes. The project concept of the degree programme is in line with the institutional integration of industry to increase attractiveness in accordance with the faculty development plan. Local industries form motivated projects in line with the sustainability requirements of the degree programme.

- **Qualification objectives/learning outcomes and subject focus**

The Engineering for Sustainability programme mainly covers topics from mechanical engineering, particularly from the perspective of sustainability. Students earn a Bachelor

⁶The whole content for the subsection was taken from [27] and translated to english

of Science degree, which is entirely competence-oriented. The aim is to provide future engineers with the necessary skills to enable them to familiarise themselves with topics outside their field of expertise and thus engage in lifelong learning, against the backdrop of open or changing requirement profiles. This will enable them to solve as yet unknown technical challenges and sustainably fulfil the associated social responsibility.

- **Special features of the programme**

The programme is designed in accordance with the recommendations of the Science Council for a sustainable structure of study and teaching from May 2022. Accordingly, the programme is project-oriented and attaches great importance to academic mentoring. It breaks with conventional structures and, through an efficient block model, the abolition of traditional lecture and exercise scenarios, a specialist department structure called a Centre of Knowledge, which is headed by the professors of the programme, and blended learning self-study formats, it provides scope for intensive project work. Students put together their own individual study programmes with the support of the teaching staff. Modules are at least 80% formulated in terms of the skills to be acquired, with the remaining 20% allowing for further individualisation of learning objectives. The requirements resulting from the overall learning objectives of a module are agreed at the beginning of the semester as part of the academic mentoring programme, continuously reflected upon during the semester and assessed as part of a 'continuous assessment' process.

In summary, the following key features of the programme can be summarised as follows:

- Academic mentoring: Consistent implementation of the recommendations of the Science Council on supporting and advising students
- Project orientation: Application-oriented teaching of course content – including the fundamentals – starting in the first semester
- Competence orientation: Focus on skills rather than knowledge through a problem-based teaching approach
- Sustainability competence:
- Meaningful, future-oriented programme approach through the integration of sustainability aspects into all modules
- New time models: Block formation in centres instead of classic course structures with separate lectures, exercises and practicals
- Integral approach: Smooth transitions between theoretical and practical components through...
- New Work: ...laboratory-like teaching and learning settings for independent and cooperative learning
- Individualisation: Selection of learning paths and co-design to address strengths and weaknesses
- Self-study: Development of personal, lifelong learning skills through blended learning and home office time
- New understanding of roles: Cross-cohort work with individual learning level-adapted tasks and responsibilities



Qualification objectives and level of qualification in Engineering for Sustainability

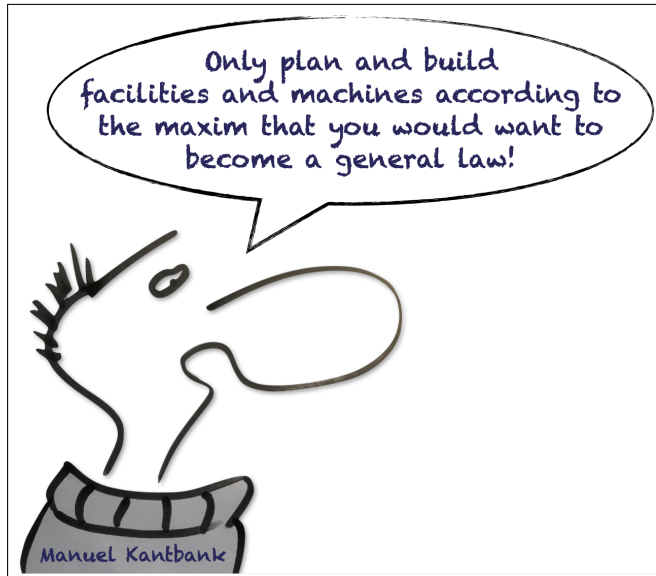


Figure 1.4: Manuel Kantbank, rumoured technical philosopher assumed to reside in the crane near the Spoy-Canal on the HSRW campus in Cleve

Sustainability stands for both the orientation and the attitude of teachers and learners. Although sustainability is now used inflationary in all kinds of contexts, for future engineers sustainability must be an irrefutable criterion in their daily work. It is something of a categorical imperative for engineers – loosely based on Immanuel Kant’s principle: *Only plan and build facilities and machines according to the maxim by which you can at the same time want them to become a universal law*. Sustainability is therefore not assigned to a ‘Centre of Knowledge’. Sustainability is a subject area in which knowledge can be imparted, but here sustainability is seen as a framework that spans the modules. It is the spirit of the entire course, the attitude with which projects are planned and implemented.

The aim of the programme is to train students and build their skills. This was al-

ready the case in the past, but in this programme, skills are the absolute top priority. There is a list of specific skills that students are expected to develop, and these are weighted differently. The list is based on relevant research findings on ‘future skills’. This enables students to acquire a programme-specific skills profile. The skills profile is shown on the final certificate. Discussions with employers indicate that future job advertisements will place significantly more emphasis on a specific skills profile than on pure specialist knowledge. At the first taxonomy level, professional competence also includes specialist knowledge, which is of course an important skill in our list, but it is flanked by other skills that are gaining in importance. All modules in the programme consistently address the overarching competencies, which are briefly listed and described below and are covered in Bloom’s taxonomy levels, taking into account different dimensions of knowledge:

- **Professional competence:** The term professional competence covers specialist knowledge and methods, as well as their application, which are necessary for accomplishing subject-specific tasks. Specialist knowledge should correspond to the current state of research.
- **Methodological competence:** In contrast to the above-mentioned professional competence, general methodological competence refers to knowledge, skills and abilities that can be applied independently of the subject matter and enable new and complex tasks and problems to be tackled independently and flexibly. These are prerequisites for the selection, planning and implementation of meaningful solution strategies. General method-

ological competence includes, for example, problem-solving skills, transferability, abstract and networked thinking, and analytical skills. Confident use of computers and foreign language skills can also be classified under the heading of methodological competence.

- **Sustainability competence:** Sustainability skills refer to the skills taught in education for sustainable development. Education for sustainable development focuses strongly on enabling individuals to actively participate in the analysis and evaluation of unsustainable development processes, to orient themselves towards sustainability criteria in their own lives, and to initiate sustainable development processes together with others, both locally and globally. Accordingly, sustainability competence serves specifically to gain shaping competence and enables decisions that, through the application of knowledge about sustainable development, supported by contemporary analyses and future studies, bring ecological, economic and social aspects into a mutual balance.
- **Personnel competence:** Personnel competence refers to the ability and willingness to develop oneself and to develop one's own talents, motivation and willingness to perform, as well as the development of specific attitudes and an individual personality.
- **Social competence:** Social competence refers to knowledge, abilities and skills relating to communication, cooperation and conflicts in intra- and intercultural contexts. It enables people to act appropriately in relationships with others and to achieve individual and shared goals.

Onboarding Semester - Welcome to the Club!

The introductory phase of study is considered particularly important, especially in degree programmes that accept a very diverse and international student body. The orientation semester is of particular importance in this regard, as it serves to socialise students into the degree programme and foster a sense of identification with the campus, the faculty and the university as an institution, and therefore requires a high level of attention. Furthermore, structured free time should give students the opportunity to synchronise their own preferences with the range of courses on offer. This takes the form of orientation programmes grouped into blocks under the interdisciplinary responsibility of the Centre of Knowledge. The 3-week blocks prepare students for the temporal project structure of the degree programme. Specifically, the 17 (15+2) weeks of the semester are divided as follows:

- **Module 2203 - Onboarding and Orientation:**

Good integration into the social environment of a university is essential for successful academic socialisation and should not be taken for granted. The first three weeks are clearly devoted to this preparatory integration, with students organised into tutor groups, establishing close contact with their fellow students from higher semesters and the responsible academic mentors, and exploring the professional field through excursions, museum visits, and visits to interpretation centers. This direct interaction also has the task of exploring possible paths through the program and their results for the students' own career profiles, taking into account individualised interests and strengths to date. The students are



prepared for the freedom of choice available to them through comprehensive information provided by the mentors.

- **Module 2202 - Working in Laboratories and Technical Centre:**

Students will work on real technical installations at the respective centers in order to develop project planning skills at this early stage. Ideally, a technical application case can be found that reflects the technical expertise of several centers in a joint project. Designing such a joint project is the task of the participating teachers and can also be seen as an opportunity to promote cooperation between the centers. The module thus gives students the opportunity to gain insights into possible areas of specialisation.

- **Period of reflection I:**

By integrating the examination period (see above) at the end of the lecture period, continuous feedback discussions with the academic mentor are possible. These reflection phases are intermediate time resources for early independent development of content and reflection by students on what they have experienced so far. During this week, tutors and mentors are available for discussions, and laboratories and technical facilities are open to students as far as possible. No further requirements are imposed in order to give students the opportunity to organise themselves freely.

- **Module 2201 - Fundamentals in Personal and Social Competencies:**

The aim of this module is to support students in developing and expanding their personal and social skills and to sensitise them to the importance of continuous, lifelong personal development in the area of key competencies. Core competencies that form an important basis for employability and success in any future professional context, whether as an individual or as a member of a team or organisation. The focus is on experiential learning, with students learning to work on real projects in teams. All competencies in this area are addressed and integrated in the module in one form or another form as a basis in the module. However, the focus of this module is on the following areas: communication skills (including conflict management, feedback, introduction to negotiation and presentation), cooperation skills, with a particular focus on team skills (team building and team roles), diversity and intercultural skills (raising awareness for the constructive handling of any form of differences).

- **Period of reflection II:**

See above.

- **Module 2204 - Abstraction and logical reasoning:**

Each center has its own methodological approach to technical issues. The respective strategies are applied in connection with the learning tools established in the first and second blocks in order to teach the basic nature of modelling, with a particular focus on the representation of real technical systems in computable abstractions. The gap in reasoning for complex mathematical content, which is a constant feature of further study, is closed here directly through explicit connection to the previously experienced practical content. This provides a direct introduction to the fundamental nature of modelling, with a particular focus on the representation of real technical systems in computable abstractions. The gap in reasoning for complex mathematical content, which is a constant feature of further study, is closed here directly through explicit linkage to the practical content previously

experienced. This provides the necessary acceptance for deeper, more demanding, and more strenuous familiarisation with complex problems right from the start.

- **Module 2205 - Sustainable and Independent Group Project:**

The aim of this final part of the orientation semester is to produce a tangible technological product in the literal sense of the word. Its functionality is to be demonstrated in a joint final event. In addition to the purely technical implementation, which includes sketches, drawings, calculations, and construction, the focus here is particularly on team building and experiencing the joy and fun of working together.

Project Semesters 2 to 5

As an educational and didactic approach, those involved in the program consistently focus on project-based learning. Projects are the central modules in the curriculum and are carried out in the "Centers of Knowledge." The content of the Centers corresponds to the areas of specialisation of the professorships. The various disciplines are located there, e.g., the Center for Strength and Simulation, the Center for Digital Product Creation, Centre for Biological Transformation, and Centre for Science Communication. This list illustrates the interdisciplinary and holistic approach of the program concept. A project is primarily an engineering-oriented task that students should solve in teams (or independently in individual parts). However, each project reflects the competence profile of the program. Of course, this cannot be done one-to-one in every discipline, but it will be the case for the projects as a whole.



Figure 1.5: Exemplary course of study from a student's perspective, with CP shares for the Project Center growing over the semesters and a corresponding focus on a few disciplines. CP shares for the Project Center growing over the semesters and a corresponding focus on a few disciplines, taken from [27]

After an orientation phase (1st semester), students begin projects in the 2nd semester that become more challenging and interdisciplinary in the 3rd, 4th, and 5th semesters. Specifically, this



means that projects are offered in cooperation with several Centers of Knowledge. When choosing projects, students have the freedom to acquire the professional competence(s) that appeal to them most. The plural form in brackets shows that students can develop into experts in one discipline on the one hand, or into generalists with expertise in several disciplines on the other. In the 6th semester, In the 6th semester, students further deepen their specialist knowledge in an internship or a research semester and complete their studies in the 7th semester with a bachelor's thesis.

Based on the experiences and feedback from the jointly completed first orientation semester (see below), students make a preliminary selection of their own specialisation in consultation with their respective mentor. The results and experiences of the students are taken into account for analysis. Students justify their selection by means of a concept paper, which is confirmed by the mentor. From the eight available areas, five are selected for further specialisation in the second semester. The results and experiences of the student are taken into account for analysis. Students justify their selection by means of a concept paper, which is confirmed by the mentor. From the eight areas available, five are selected in the second semester for further specialisation. In the third semester, after similar feedback from the mentor, a selection of four specialisations is made based on inclination and goals. It is possible to change from the initial specialisation to a new one once, whereby the student is required to independently develop the entry requirements for the second stage of the project phase (3rd semester) as specified by the responsible teaching staff and pass an examination. Similarly, further specialisation in three areas takes place in the fourth semester and in two areas in the fifth semester.

Walking (nearly) without a Net - The concluding Semester 6 and 7

The sixth semester is followed by an internship that covers the two areas of the fifth semester in an interdisciplinary manner. An independent final thesis in one or both areas of specialisation completes the programme in the seventh semester.

Exercise 3:

Summarise the most important aspects of the degree programme in 10-20 keywords. What needs to be clarified?

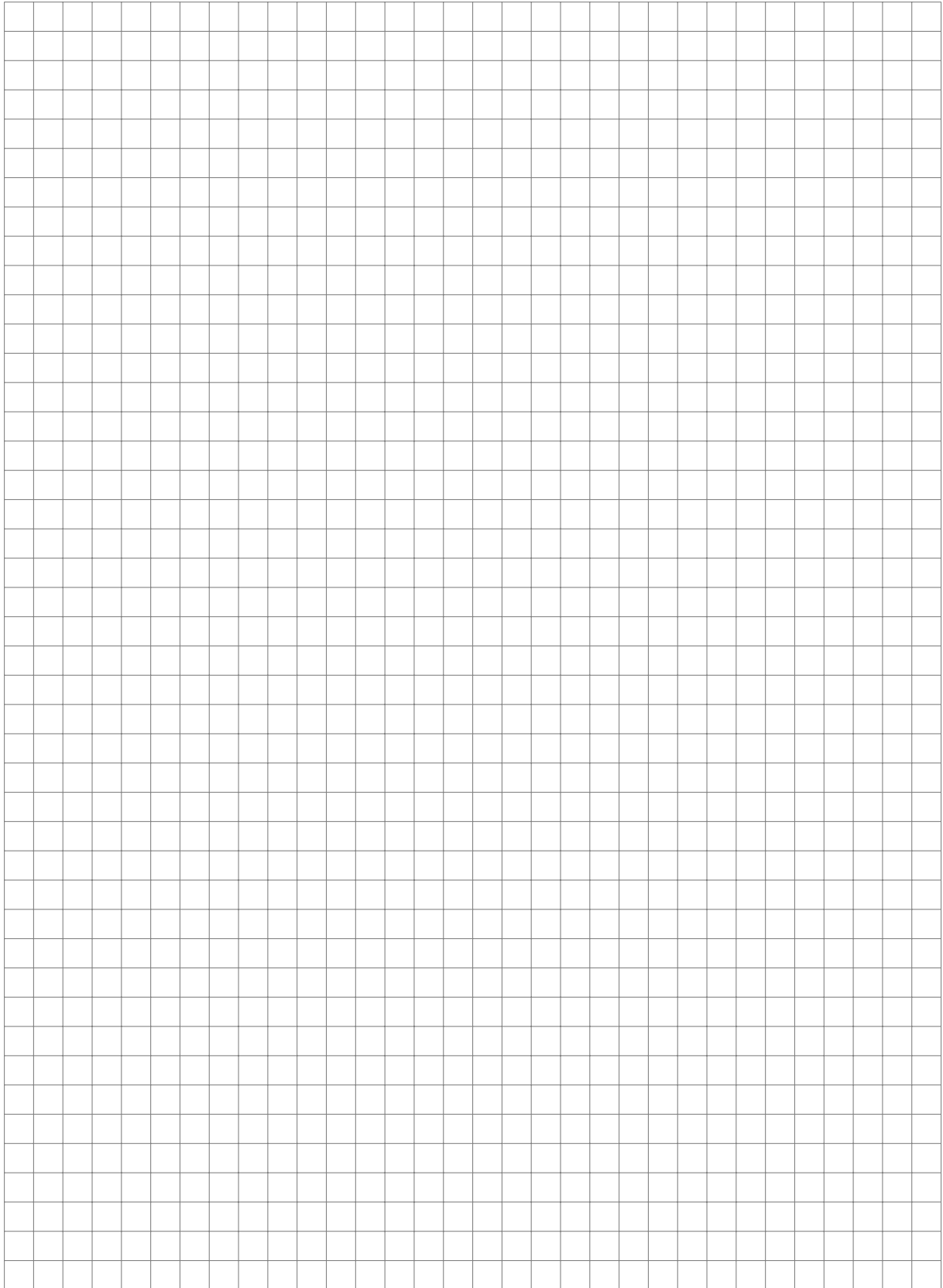
2.3 Motivational Aspects

Exercise 4:

Why did you enrol in the study programme Engineering for Sustainability? Try also to describe the person that you will be when having finished your academic career and what your impact on society will be. - You will read that definitely at some later point in your life, so why not putting that into a letter to yourself? Remember that you will be stuck with yourself throughout your whole life - so be kind ;-).

Dear future me,

This image shows a full page of blank graph paper. The grid consists of thin, light gray horizontal and vertical lines that intersect to form small squares across the entire surface. There are no margins, text, or other markings on the paper.



2.4 Some Tools for Technology Assessment

Warming up - What will the weather be when it is all over?

The title of this subsection is clearly misleading - there is a clear distinguishment between weather and climate (which is exactly defined how?). Climate change (for more information see e.g. [41]) is addressed often as one of the main aspects when discussing sustainability. The exact global warming is certainly difficult to predict (and maintaining a scientific view, scepticism is a major feature to hold on to), but the rise of global averaged temperature is usually treated as a fact. A rough prediction is shown in fig. 1.6, the original caption of the publication [24], being a takeout of the full report [25] was stating 'Human-induced warming reached approximately 1°C above pre-industrial levels in 2017. At the present rate, global temperatures would reach 1.5°C around 2040. Stylised 1.5°C pathway shown here involves emission reductions beginning immediately, and CO2 emissions reaching zero by 2055.' The publication was made by the ipcc - the *Intergovernmental Panel on Climate Change* which is the United Nations body for assessing the science related to climate change.

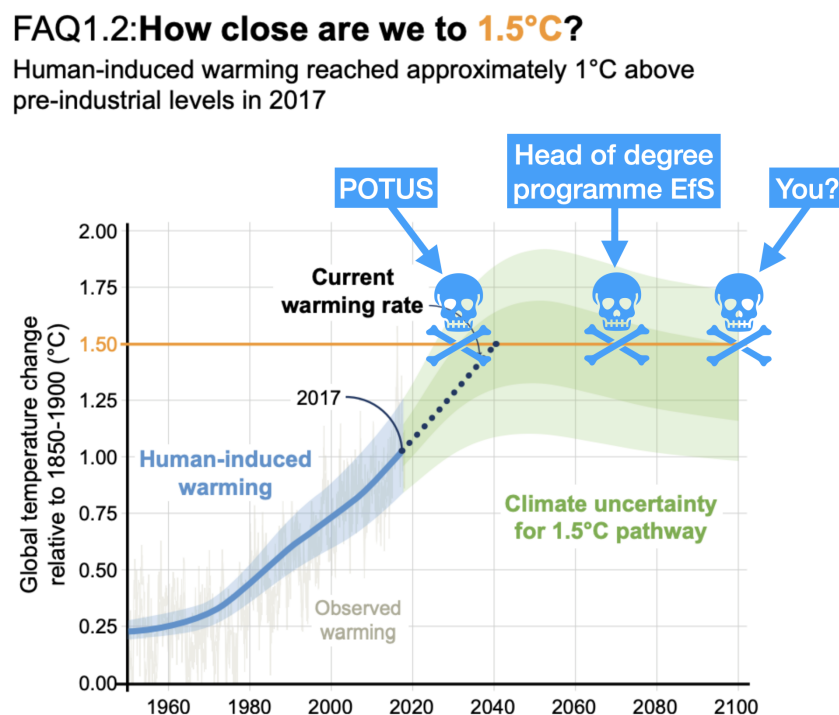


Figure 1.6: Global temperature change, original figure taken from [24], skulls and blue text added by the author, dates of placement POTUS and HoDP-EfS taken from [60]

The figure was modified by the author in terms of adding predicted life expectancies from [60] for the acting POTUS (reported to decrease efforts on preventing climate change, [73], also making efforts to 'recontextualise' phenomena of global warming, see [10]) and for the head of the EfS study programme (actively trying to increase efforts on preventing climate change) as of 2025.

**Exercise 5:**

Using the life expectancy predictor [60], what is your personal 'ending point'? Mark that point in fig. 1.6. Also, while going through the life expectancy questions, what is your potential intention on influencing the time set by your way of life?

Exercise 6:

Does it matter for you what comes afterwards? For which time period do you think it makes sense minimising your own impact by your chosen lifestyle? Decades? Centuries? Millennia? At some point our efforts have to end as our biosphere will supposedly decline inevitably in the next 1 to 2.8 Gyr [51], [50]... Make a statement on a reasonable 'ending point of efforts'.

The Brundtland Definition of Sustainability

Blessed is he who plants trees under whose shade he will never sit., although the exact quotation is subject to several discussions [53], origins of the proverb can at least be backtracked to the roman empire to a quote by Marcus Tullius Cicero⁷. The thought of an alturistic caretaking for future generations is not an invention of the last decades and throughout history seen as an indicator for a well functioning society.

In the official pages of the german parliament [15], the translated term „Nachhaltigkeit“ is rooted back into forestry: *'Those who only fell as many trees as can grow back ensure that the forest remains available for future use and retains its value in the long term.'*⁸

A more up to date definition is given by the the United Nations Brundtland Commission for whom *sustainability* is *'meeting the needs of the present without compromising the ability of future generations to meet their own needs.'* [12]

Exercise 7:

Check the original source [12]. Where exactly is the definition stated in what form? Check the definitions against other definitions you find. Which definition fits your own understanding best (name the correct reference)? If you do not find a definition that perfectly suits your own perception of sustainability state your own!

⁷serit arbores, quae alteri saeclo prosint, according to [53]

⁸own translation of the original quote: „Wer nur so viele Bäume fällt, wie nachwachsen können, sorgt dafür, dass der Wald für die künftige Nutzung zur Verfügung steht und auf Dauer seinen Wert behält“

The Sustainable Development Goals of the United Nations

Sustainability is much more than performing a simple (?) energy balance for a brown coal power plant and involves a much broader view on the topic. A well known effort to summarise sustainable targets into a finite set of goals is given by the *sustainable development goals of the united nations* (SDG), as depicted in a pyramid structure in fig. 1.7.



Figure 1.7: Sustainable Development Goals Pyramid, taken from [70]

An easy accessible overview on the 17 goals is given by the UN themselves: <https://www.un.org/sustainabledevelopment/sustainable-development-goals/>

Exercise 8:

Clearly some of the development goals are interacting with each other - can you identify similar goals? If so, group them and find superordinate terms (besides the levels in the pyramid).



The Triple Bottom Line

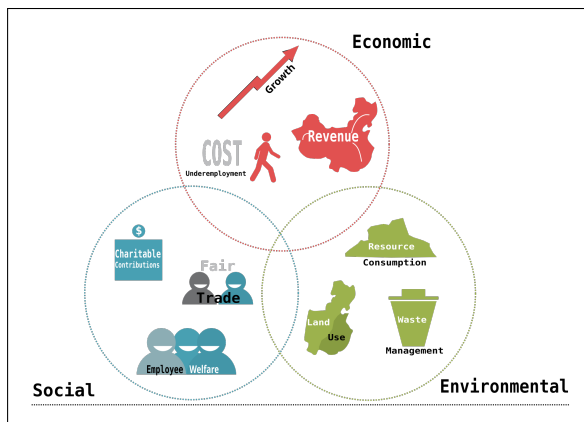


Figure 1.8: A graphic describing the some of the areas of study that the Triple Bottom Line framework works with, from [11]

In economical projects, the final judgment on the success is mostly made by monetary profit, the so called bottom line calculation. In sustainability focussed projects, there is not only one, but (at least) three bottom lines which should all have a positive balance:

- planet (environmental)
- profit (economic)
- people (social)

These goals are usually referred to as *triple bottom line*, including an enhanced understanding of profit as depicted in fig. 1.8.

An overview of exemplary aspects to include is given in fig. 1.8.

Table 1.1: triple bottom line, content taken from [39], own table

Planet	Profit	People
Ecology	Corporate	Social Aspects
Biodiversity	Region	Participation
Resources	Global	Decision making
Emissions	Wealth	Community
Imissions	Welfare	Justice
Greenhouse	Education	Future Generations
Σ	Σ	Σ

Exercise 9:

Compare the entries in fig. 1.8 to the SDGs given in fig. 1.7, are these somehow related? Assign every SDG to one of the pillars in tab.1.2

Table 1.2: triple bottom line and SDGs

Planet SDGs	Profit SDGs	People SDGs
Σ	Σ	Σ

Exercise 10:

You already found out that the SDGs are interrelated and every SDG has an impact on the others, can you imagine how high the impact of every SDG on all of the others is? Fill out the respective impact a SDG (written in a line) has on all the others (named in the respective column) in tab.1.3

Table 1.3

SDG ↓ impacts SDG → [(H)igh/(M)id/(L)ow]	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1	x																
2		x															
3			x														
4				x													
5					x												
6						x											
7							x										
8								x									
9									x								
10										x							
11											x						
12												x					
13													x				
14														x			
15															x		
16																x	
17																	x

Exercise 11:

Not only the SDGs themselves, also there is an influence between geographical regions. Interpret the impact of the situation in surrounding countries on the EU, see fig. 1.10.

SDGs IMPACT FACTORS OF NEIGHBOURHOOD COUNTRIES ON THE EUROPEAN UNION																	
SDG	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Russia	4	3	3	3	4	8	10	10	6	6	6	7	4	3	7	10	5
Ukraine	10	3	8	7	8	8	3	10	3	10	6	10	3	2	4	10	8
Turkey	10	4	8	7	7	8	7	10	4	10	6	6	3	4	4	10	8
Egypt	7	4	5	5	5	8	5	5	2	5	4	3	3	4	3	7	5
Iran	7	4	6	8	5	5	3	5	4	5	4	3	3	2	3	10	5
Israel	8	5	8	8	5	10	10	10	5	8	7	3	5	4	3	10	10
Palestine																	

SOURCE: BASEL INSTITUTE OF COMMONS AND ECONOMICS 2018

Figure 1.9: SDG Impacts on the EU, from [18]



Exercise 12:

The named goals have a different importance or relevance for people from different countries. What do you think is the impact of all the respective SDG on your home country? Use the following tool for clarification and also helping to provide data for the UN <https://trustyourplace.com/?lang=en&country=Germany> Compare your result to the impacts of the countries as classified by [18] depicted in fig. 1.10

The European Union SDGs Impact Factors																	
SDGs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Austria	1	1	1	1	1	1	1	1	1	1	1	3	2	1	2	3	1
Belgium	1	1	1	1	1	3	5	2	2	2	2	3	3	3	2	2	1
Bulgaria	5	3	4	4	3	3	3	4	6	6	5	5	3	3	3	8	1
Croatia	5	1	3	4	3	3	4	4	6	4	4	3	2	4	2	3	1
Cyprus	4	1	2	2	2	4	4	4	6	4	4	4	2	4	3	10	2

Figure 1.10: Sustainable Development Impacts, from [18]

Exercise 13:

Again, reflecting on your home country. Which SDGs do you think are well taken care of on your country and in which is your country performing worst? After fixation, check your evaluation by comparison with the sustainable development report: <https://dashboards.sdindex.org/map/>

Exercise 14:

For all countries, a fulfilment of the SDGs still requires a lot of work. Every SDGs is subdivided into different indicators for which the performance is rated, e.g. the lack of performance for SDG 4 (education) in Germany is mainly caused by the weak performing indicators 'Variation in mathematics performance explained by socio-economic status' and 'Underachievers in mathematics' (that is we we try to counteract on that by our module 2204 *Abstraction and logical reasoning* in the first semester). Find out which fields of action in the different SDGs require the most urgent attention by checking your countries entry in the *Sustainable Development Report* [47].

Contextualisation by Hermeneutics

When assessing technology, the evaluating person is in his or her judgement limited and influenced by previous experiences, inner predisposition, personal goals, motivations, behavioural codices etc.. Due to limitations of the human mind and individual perceptions, no two individuals will share the same prerequisites when performing an evaluation. As a consequence, judgements of two persons of the same object or process might (or will) differ. An awareness of the non achievable objectivity can support a reduction of the consequences of the personal inherent subjectivity when making a statement, or at least put the statement into perspective.

Hermeneutics, initially a concept for understanding and interpreting religious, historical and philosophical texts, is following the fact that a text written in a given historical context had a meaning (which most of the readers were aware of, as these had all the 'common knowledge' of their given time) which is exceeding the amount of information that can be preserved by the pure words in which the information is condensed (which also change their meaning over time). The named 'context' can also be a location-specific context⁹ or a context that is referencing information that is only available inside of a given 'social-bubble'.

Conceptual alignment:

1. Choose a technical product that you think is very important to a given group in your home-country. State the context making this product valuable.
2. Change the local context of that product and virtually place it in the hands of another group (e.g. a group of members of a given vocation, hobby, age etc.) in your country. Again state the context that influences the value of the given product. How did the value change in comparison to its initial environment? What is changed when changing the country defining the context?
3. Change the temporal context, place your chosen product in chosen local coordinates in a different timeframe, how does the reception of the product change in -10, -100, -1000 and +10, +100, +1000 years?
4. As the concept of hermeneutics is developed in the context of philosophy and Humanities, of what relevance is the method for the assessment of sustainability of technical products? For getting the discussion started, a review of an AI-generated statement is done:

Prompt[55]: What relevance does hermeneutics have for the assessment of the sustainability of a technical product?

a) Basic idea: Hermeneutics

Hermeneutics is the theory of understanding and interpreting contexts of meaning. Originally developed in theology and literary studies, modern hermeneutics (e.g., Gadamer, Ricoeur) is about understanding within contexts: everything only gains meaning within the horizon of history, culture, language, and practice. An object

⁹in reference to art e.g. in 'Das Werk, aufbewahrt in der Sammlung oder dem Depot, ist aus dem Geschehen seiner Wahrheit herausgenommen. [...] Es steht da als ein gegenständliches Ding unter anderen. In solcher Aufbewahrung stirbt das Werk.' [35] S.14



or text “speaks” only when we interpret it—when we relate it to ourselves and our world-understanding.

b) Application to technical products

For technical products this means: products are not just physical things, but always embedded in cultural, social, and ecological contexts. Their “sustainability” is not an objective, purely technical value, but an interpretive attribution depending on horizons of understanding (e.g., what a society defines as “sustainable”).

c) Hermeneutical relevance for sustainability assessment

- i. **Contextualization.** A product is sustainable or not depending on the context in which it is viewed. For example, an electric car may appear sustainable if only its operation is considered; less so if battery production and disposal are included. Hermeneutics sharpens awareness that evaluation depends on the horizon of interpretation.
- ii. **Interpretation of life cycles.** Life cycle analyses provide data, but the meaning of this data (e.g., CO₂ equivalents, resource consumption) must be interpreted. Hermeneutics asks: How do we understand these numbers in light of our values and goals?
- iii. **Understanding user and cultural relations.** Products are sustainable only if they are socially accepted and meaningfully used. Hermeneutics asks: What meanings do users ascribe to the product? A long-lasting product that no one wants to use is practically not sustainable.
- iv. **Reflection on value concepts.** Heidegger shows that the notion of “value” often arises when something has been removed from its living context. Applied here: sustainability is not just a “value label,” but a relational event that must be understood and continually reinterpreted.

d) Practical implications

- **Methodological:** Assessing sustainability cannot be purely technical; it must include hermeneutical reflection (e.g., stakeholder inquiries, cultural contexts, future horizons).
- **Ethical:** Hermeneutics makes clear that every sustainability assessment is an act of interpretation and value-setting— and therefore never entirely neutral.
- **Transdisciplinary:** Engineers need dialogue with the social and cultural sciences in order to grasp the meaning of sustainability.

e) Summary Hermeneutics shows that assessing the sustainability of a technical product is not a purely objective, scientific procedure, but an interpretive process that includes cultural meanings, values, and contexts. It provides the critical awareness that “sustainability” must not only be calculated but also understood.

Exercise 15:

Which statements do you think are the most important for our future goal to develop sustainable technical progress? Give reason.

Iterative Contextualisation using the Hermeneutic Cycle

What is somewhat trivial in terms of engineering, that the functional requirement of a part is deduced from the purpose of the machine it is comprised of and, vice versa, the machine earns its functionality from the design of a single part and all their interactions in an ensemble, in the context of epistemological Hermeneutics it is stated *Jedes Einzelne wird aus dem Ganzen verstanden und das Ganze aus dem Einzelnen. Dieses Wechselverhältnis bildet den eigentlichen hermeneutischen Zirkel.* [58], p.75. The so called *Hermeneutic Cycle*, sketching the main thought of the given quote is shown in fig. 1.11.

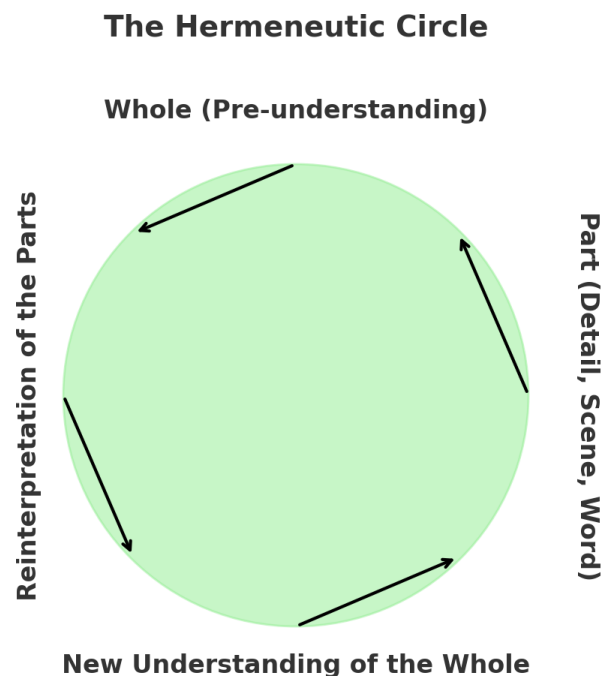


Figure 1.11: Hermeneutic cycle, [58]

Despite the process of acquiring knowledge in general, that is also a process that can be found in technological progress of inventions themselves, e.g. the more knowledge one acquires on how an engine works as an energetic-entropic system (e.g. by advancing in thermodynamics), the better the shape of the combustion chamber and piston can be designed, also the more advanced the shape and material of the piston gets (by engineering design and material sciences), the better the technical system gets (and also the understanding of the system) - iterating from the birth of the invention to the state of technical advancement that we presently have. The same applies to the assessment of the sustainability of technical inventions, process or even companies in



general: Our initial rough understanding is a prerequisite to understand the functionality of the parts, whereas a better understanding of the parts increases our understanding of the assessed entity as a whole, again leading to a better understanding of the parts. That iteration is repeated until a (to be defined) threshold of accuracy/understanding/technical advancement is made.

Conceptual alignment:

1. Name a technical system/invention undergoing the iterative understanding cycle at present. Clearly name the system and the parts and formulate several cycles of advancement.
2. In terms of judging the sustainability of a technical entity, also the judgment is influenced by the knowledge on the components. This includes the purely technical features of the entity and also the socio-economic constraints under which these were made, e.g. knowledge on child-labour involved in the manufacturing process of an otherwise eco-friendly shoe will change our judgement. Further a technical entity is also a part of a more complex interaction-web in society - which also change their values, e.g. shown in the perception of nuclear fission in the changed 'surroundings' of *environmental protection goals* in the 80s on the one and and *climate change prevention* at the present day. Find one example respectively for...
 - a) a technical entity where you changed your judgement (feasibility, sustainability, practicality) by increased knowledge on how the parts it consists of works.
 - b) a technical invention where changing conditions in society change the perception of the technology.

2.5 Preparing for the field trip - General Overview on the Company

Check the homepage of the company at hand for general information. Also take other sources of informations into account (provided material, www, library). Use the following questions for orientation, add questions as desired.

Exercise 16:

1. In which business sector is the company located? What is its size (SMU or large company) and what is the companies legal form (AG, KGaA, SE , GmbH)?
2. Which are the direct competing companies? Where are these located?
3. Which products are provided for which customers? What are the special requirements of the customers and how is that addressed in product development?
4. What resources are mostly used for the production of the products? How are the products manufactured? Which machines and which personnel is needed in the company?



2.6 Preparing for the field trip - Sustainability of the Company

Let us address the perception of the sustainability of the company of the upcoming field trip, apply the general concept of Hermeneutics to explicitly raise the awareness of the iterative process of our judgement.

Exercise 17:

Assess the companies sustainability step-wise, using the subsequent questions as a guideline. If you have better measures for a sustainability rating use these. Take key-point notes on the major aspects to share your insights with your colleagues.

1. Pre-evaluate the sustainability of the company on your present informations, summarised in a singular sentence.
2. Check if sustainability is explicitly addressed in the information that is provided by the company. What proof is given for the mentioned sustainability? Are certificates mentioned (e.g. ISO 14001, ISO 50001, *Blauer Engel* or similar - what is the relevance of these certificates?) Are specific projects named and explained?
3. Do a research on the interfaces of the company, where do the (supposedly) get their parts from? Where are these manufactured? What materials are used? Where and how are these materials (usually) harvested?
4. Check the customers of the company - what is their respective approach of sustainability? (Is that important for rating the companies sustainability)?
5. Check other activities besides the technical production process itself; is the company involved in further activities to support the broader variety of sustainability (peace-making measures, diversity campaigns, BVNU, GKV code of conduct etc.). Try to find indicators for serious commitment or rather utilisation of the memberships as marketing measure only.
6. Check reports on sustainability, if available (statistics on energy usage and mix, EU 2022/2464 (CSRD), EU 2014/95 (NFRD) etc.)

Exercise 18:

Based on your notes on the company and its sustainability, rate the companies effort considering the SDGs. Use the given matrix in tab.1.4 for guidance, add categories as needed and specify and justify your rating.

Table 1.4: Action impact (high, medium, low) matrix for company policies

1. Inform holistic 2. Pre-assess 3. Find examples 4. Find proof 5. Define indicators 6. Rate indicators 7. Re-assess 8. Repeat	Product manu- factured	Supply chains used	General product- cycle and lifetime consider- ations (disposal/ recycling)	Energy investment for products and sources of energy	Cross- company activities	Community and charity commitment	Employees and other stakeholders	Behavioural codices and company ethics	additional criterion 1:	additional criterion 2:	additional criterion 3:
Hunger and Poverty (1,2)											
Human Demand (7,3,6)											
Nature (14,15,13)											
Education and Technology (4,9,12)											
Socioeconomic Development (8,11,17)											
Justice (5,10,16)											

Explanation and justification of rating:

Exercise 19:

Considering your rating, which information was insufficient for a rating beyond doubt? Prepare questions to clarify on the upcoming field trip:



2.7 While Being on the Field Trip / After the Field Trip

Exercise 20:

- Get an answer to your open questions formulated above.
- Reevaluate your judgement on the present sustainability of the company using tab.1.4.
- Imagine possible actions of the company to increase your judgement on the sustainability of the company as a whole.

Sustainable project at the company.

Exercise 21:

Briefly describe the sustainable project shown at the field trip.

Exercise 22:

Evaluate the project using tab.1.5 and give reason for your evaluation.

Table 1.5: Action impact (high, medium, low) matrix for projects, in reference to [38], action areas modified

1. Identify Status 2. Define Focus Areas 3. Define Goals 4. Define Actions 5. Track actions and results 6. Evaluate and assess 7. Repeat	Goals	Adjacent projects	Impact on external stakeholders	Supply-chain	Impact on employees	Aftermarket activities			
Hunger and Poverty (1,2)									
Human Demand (7,3,6)									
Nature (14,15,13)									
Education and Technology (4,9,12)									
Socioeconomic Development (8,11,17)									
Justice (5,10,16)									

2 Behavioural Codices for Engineers

1 Ethical Standards in Companies - A Field Study

(content will be replaced)

1.1 Sustainability, Normative Competencies and Ethics

Der Leonaridische Eid *Eine jede Ingenieurausbildung muss sicherstellen, dass Ingenieurinnen und Ingenieure darin gebildet werden, ihre fachlich basierten Gestaltungsmöglichkeiten einzubringen, um ihrer hohen Gestaltungsverantwortung anhand der Prinzipien*

- *der ethischen Legitimierbarkeit,*
- *der Nachhaltigkeit und*
- *der gesellschaftlichen Kontrollierbarkeit*

gerecht zu werden. [29]

Conceptual alignment:

1. Discuss on the importance of the mentioned principles for
 - a) engineers in general
 - b) an engineer for sustainability
2. If you disagree with the importance: What should be done instead? If you agree with the importance: How do you wish these principles should be included in the curriculum?

1.2 The Usage of Technology - An Engineers Responsibility?

One of the wonderful things about philosophy is that it is possible to have deep disagreements and offer arguments against one another's views, while doing this in a way that is friendly and respectful toward all those involved in the discussion. (...) On social media platforms, for example, one sometimes sees people expressing regret that discussions that start on such platforms so easily escalate into non-productive, angry exchanges.(...) That raises another issue discussed in technology ethics:

is it unethical to create technologies that may lead some people to act in morally problematic ways?

[49], p.xiii.



Conceptual alignment:

1. List technologies that have lead (or currently) people to ‘morally problematic ways‘ and summarise the potentially unethical usage of the named technology (briefly, bullet-point like)
2. Do the same as stated above for future technologies.

3 Engineering Disciplines and Competencies

1 Engineering Skills in Companies - Self Assessment

Needed skills are widespread for an engineer, especially in the context of sustainability. Also the most important skills mentioned are depending on whom you ask - for a reason - engineers are employed in vastly different working environments with an equally vast amount of different skills used on various tasks. In the end you have to decide by yourself which skills to level and which to ignore. There is no correct answer, and the easy way developing skills is by simply using them. If you have a distinctive picture of the person you want to be when finishing your studies, a little bit planned usage of skills to enhance those fitting to your future plans is reasonable. To get a better overview on the skills available for you to choose from, we will have a look at some of those here.

1.1 Self Assessment Questions on the Attitude when becoming an Engineer of Sustainability

The questions presented in the following are planned to be used in the upcoming years as OSA for the study course; so please fill them out truthfully and conscientiously. Also be prepared to give a feedback on the feasibility for the study course at hand. At the end of the section you are asked for a quick statement on how to improve the questions for the upcoming generation of students, e.g. one word/bullet point on what to keep/drop/try in the next version.



Technical Approach / Mathematical-Physical Modelling

Please mark your (dis-)agreement on the given scale under the respective question (1=total disagreement / 10 = full agreement)

1. I feel at home in STEM subjects.

(total disagreement)	1	2	3	4	5	6	7	8	9	10	(full agreement)
----------------------	---	---	---	---	---	---	---	---	---	----	------------------

2. I am persistent in finding solutions to complex problems. Complicated and challenging constraints motivate me.

(total disagreement)	1	2	3	4	5	6	7	8	9	10	(full agreement)
----------------------	---	---	---	---	---	---	---	---	---	----	------------------

3. I am very interested in current technical developments and take the initiative to obtain specialist books and other learning materials.

(total disagreement)	1	2	3	4	5	6	7	8	9	10	(full agreement)
----------------------	---	---	---	---	---	---	---	---	---	----	------------------

4. I enjoy working with computers, have already carried out computer-aided simulations myself, and written small programs.

(total disagreement)	1	2	3	4	5	6	7	8	9	10	(full agreement)
----------------------	---	---	---	---	---	---	---	---	---	----	------------------

5. I like to familiarise myself with the operation of new technical equipment and discover its functions.

(total disagreement)	1	2	3	4	5	6	7	8	9	10	(full agreement)
----------------------	---	---	---	---	---	---	---	---	---	----	------------------

6. I am not afraid to get my hands dirty with machines or to perform the necessary technical calculations in great depth.

(total disagreement)	1	2	3	4	5	6	7	8	9	10	(full agreement)
----------------------	---	---	---	---	---	---	---	---	---	----	------------------

7. I am conscientious when it comes to creating documents (documentation, verification calculations, drawings).

(total disagreement)	1	2	3	4	5	6	7	8	9	10	(full agreement)
----------------------	---	---	---	---	---	---	---	---	---	----	------------------

Summary: Count your total agreement score: ____/70

Self-Organization and Cooperation

Please mark your (dis-)agreement on the given scale under the respective question (1=total disagreement / 10 = full agreement)

1. I carefully prepare important decisions in my life. I gather thorough information and plan my approach systematically. I find it easy to make necessary adjustments.

(total disagreement)	1	2	3	4	5	6	7	8	9	10	(full agreement)
----------------------	---	---	---	---	---	---	---	---	---	----	------------------

2. In a working group, everyone has their role, and I am able to take on different roles as needed (expert, coordinator, mediator, etc.).

(total disagreement)	1	2	3	4	5	6	7	8	9	10	(full agreement)
----------------------	---	---	---	---	---	---	---	---	---	----	------------------

3. I plan my weekly activities in advance. Physical activities have just as fixed a time frame as learning periods, working hours, sleep, and rest.

(total disagreement)	1	2	3	4	5	6	7	8	9	10	(full agreement)
----------------------	---	---	---	---	---	---	---	---	---	----	------------------

4. If I get stuck on a problem, I actively seek support from others. Their professional expertise is more important to me than personal sympathies.

(total disagreement)	1	2	3	4	5	6	7	8	9	10	(full agreement)
----------------------	---	---	---	---	---	---	---	---	---	----	------------------

Summary: Count you total agreement score: ____/40

Additional question: My daily average work performance P in the working weeks before a deadline is best described by the following function (depending on time τ)

- | | | |
|---|--|---|
| ☐ $P = 0$ | ☐ $P = \text{const.}$ | ☐ $\mathbb{Q}_*^+ = \{ q \in \mathbb{Q} \mid q > 0 \}$ |
| ☐ $P \propto e^\tau$ | ☐ $P \propto \tau$ | ☐ $P \propto \ln \tau$ |
| ☐ other: _____ | | |

Please mark your (dis-)agreement on the given scale under the respective question (1=total disagreement / 10 = full agreement)

- | | | | | | | | | | | | |
|----------------------|---|---|---|---|---|---|---|---|---|----|------------------|
| (total disagreement) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | (full agreement) |
|----------------------|---|---|---|---|---|---|---|---|---|----|------------------|

- | | | | | | | | | | | | |
|----------------------|---|---|---|---|---|---|---|---|---|----|------------------|
| (total disagreement) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | (full agreement) |
|----------------------|---|---|---|---|---|---|---|---|---|----|------------------|

- | | | | | | | | | | | | |
|----------------------|---|---|---|---|---|---|---|---|---|----|------------------|
| (total disagreement) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | (full agreement) |
|----------------------|---|---|---|---|---|---|---|---|---|----|------------------|

- | | | | | | | | | | | | |
|----------------------|---|---|---|---|---|---|---|---|---|----|------------------|
| (total disagreement) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | (full agreement) |
|----------------------|---|---|---|---|---|---|---|---|---|----|------------------|

Additional question 1: How do you assess the importance of sustainability in technology?

[illegible][illegible]

Improving Assessment Questions on Attitude Reflecting on the questions asked, considering their purpose on evaluating the 'accuracy of fit' for the study course EfS: What is your opinion on these and what should be done for improvement? What to keep/drop/try?

1. General opinion on feasibility:

[illegible]

- ## 2. Things worth keeping:

[illegible]

- ### 3. Content to drop:

[illegible]

4. Next time, try to...

[illegible]

5. A question to add:

[illegible]

Thank you very much for your feedback



1.2 Self Assessment Questions on the Skills Needed within the Respective Center

When asked for skills needed in the study course, the heads of the respective center of knowledge had provided the following questions. Not all of them are necessarily closely related to the actual content of the center, but might give you sense on what is assumed to be most important by the persons working in the vocational field. When more profound questions arise, at best ask the persons related to the respective center (which are...?).

Centre of Knowledge Automation Engineering (AE)

Exercise 23:

Leverage: Which iron bar (as depicted in fig. 3.1) is easiest to lift the box with?



Figure 3.1: Moments of force for lifting a box

- | | |
|-------------------------------------|--|
| ☐ Iron bar A | ☐ Iron bar C |
| ☐ Iron bar B | ☐ Equally good with all three iron bars |

Exercise 24:

Glass tube: A glass tube is placed in a glass of water, see fig. 3.2. A glass sphere is attached to the top of the tube. What happens to the liquid in the tube if you hold the glass sphere in your hands for a while?



Figure 3.2: Glass tube is placed in a glass of water

- | | |
|---|--|
| ☐ The liquid in the tube rises | ☐ The liquid does not change its level |
| ☐ The liquid in the tube sinks | ☐ The liquid sinks to the water level of the vessel |

Exercise 25:

Three rain barrels of the same diameter are completely filled with water and positioned so that both the outlet and the top of the barrel are at the same height, see fig. 3.3. Which outlet produces the most water pressure?

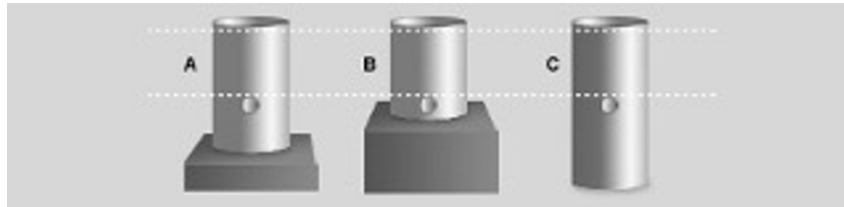


Figure 3.3: Rain barrels of the same diameter

- | | |
|---------------------------------------|---|
| ☐ From drain A | ☐ From drain C |
| ☐ From drain B | ☐ Equally strong from all outflows |

Centre of Knowledge Biological Transformation (BT)

Exercise 26:

Suppose a 65 kg person lives on a diet consisting of prime beef ($h_{PB} = 4 \text{ MJ kg}^{-1}$), honey ($h_H = 14 \cdot 10^6 \text{ J kg}^{-1}$) and corn flakes ($h_{PF} = 16 \text{ MJ kg}^{-1}$). What mass of corn flakes must this person eat per day if the diet consists of 20% by mass of beef, 20% honey and 60% corn flakes? Assume that the average metabolic power consumption is a factor 2 above the resting metabolic rate, $\Gamma = 3.6 \text{ M}^{0.75} \text{ W kg}^{-1}$.

- | | |
|----------------------------------|----------------------------------|
| ☐ 4.33 kg | ☐ 0.57 kg |
| ☐ 0.33 kg | ☐ 0.67 kg |

Exercise 27:

Which two parts of a microscope (fig. 3.4) combine to give a magnified view of a cell?



Figure 3.4: Microscope

- | | |
|--|--|
| ☐ Light source and objective lens | ☐ Eyepiece and focus knob |
| ☐ Stage and eyepiece | ☐ Eyepiece and objective lens |

Exercise 28:

Biomimetics (*Bionik* in German) is about abstracting the function of natural processes and applying the understanding obtained to the solution of engineering challenges. It is ideally suited to engineering for sustainability because:

- ☐ Bionic prostheses are performance enhancing mechanisms that enable more efficient processing of materials.
- ☐ Engineering is unable to solve the challenges facing sustainable development.
- ☐ Biological systems are inherently complex and therefore more able to adapt to changing conditions in the environment.
- ☐ Natural processes take place at room temperature and pressure, with water based chemistry, mostly at neutral pH.

Centre of Knowledge Business and Entrepreneurship (BAE)

Question loading...

Centre of Knowledge Digital Product Creation (DPC)

Exercise 29:

What is the most convenient way to manufacture the shown part (fig. 3.5)?

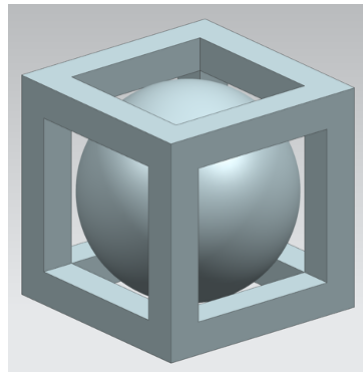


Figure 3.5: Manufactured part

- | | |
|----------------------------------|--------------------------------------|
| ☐ Milling | ☐ Forging |
| ☐ Casting | ☐ 3D printing |

Exercise 30:

How often is the marked centre block in the flow chart executed? (fig. 3.6)?

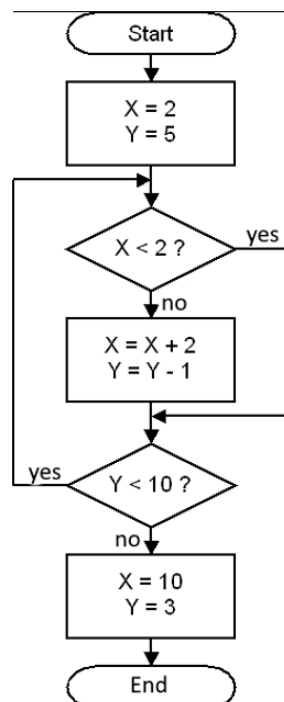


Figure 3.6: UML - scheme of a calculation

- | | |
|----------------------------|----------------------------|
| ☐ 4 | ☐ 8 |
| ☐ 5 | ☐ ∞ |

Exercise 31:



You are designing the smooth, aerodynamic outer surface of a bicycle helmet in a 3D design software. To define the guiding curves for the shape, you need to choose a spline type that offers precise control over smoothness and continuity of curvature, especially at transitions between curve segments. Which spline type is most suitable?

- ☐ Polyline – a series of straight lines connected at vertices
- ☐ Interpolated spline – a curve that passes exactly through specified points but may have uneven curvature between them
- ☐ Bézier spline – a curve controlled by endpoints and control handles, offering smooth curvature but limited to fewer control points
- ☐ B-spline (Basis spline) – a curve defined by control points that influence shape without the curve necessarily passing through them, allowing local control and smooth curvature

Exercise 32:

You performed a 3D scan of an old mechanical part to create a CAD model for reproduction. After scanning, you examine the resulting mesh and notice: The mesh contains thousands of tiny triangles, some areas have small holes, and some fine features are missing or blurred. Before you can use the scan for accurate CAD modelling or 3D printing, what is the most appropriate next step?

- ☐ Directly export the mesh to STL format and send it to the 3D printer
- ☐ Increase the scanner's resolution setting and rescan the part multiple times
- ☐ Abandon the scan and manually measure and redraw the part in CAD from scratch
- ☐ Use mesh processing tools to repair holes, smooth noisy areas, and reduce triangle count where possible

Exercise 33:

You are designing a mechanical part that will be produced using Fused Deposition Modeling (FDM) 3D printing. The design includes overhangs, thin walls, and small holes. Which design modification is most important to ensure a successful and high-quality print without excessive post-processing?

- ☐ Keep overhangs greater than 60° from the horizontal to avoid supports
- ☐ Use as many sharp internal corners as possible for precision
- ☐ Make the entire part hollow to reduce print time
- ☐ Increase wall thickness and adjust small holes to account for printer tolerance

Exercise 34:

A technical drawing (see fig. 3.7) shows a part in three views that are orthogonal to each other: front view (upper left), left-side view (upper right), and top view (bottom). Locate surface

A (which is labelled in the top view) in the front view. Which number/letter represents the requested surface?

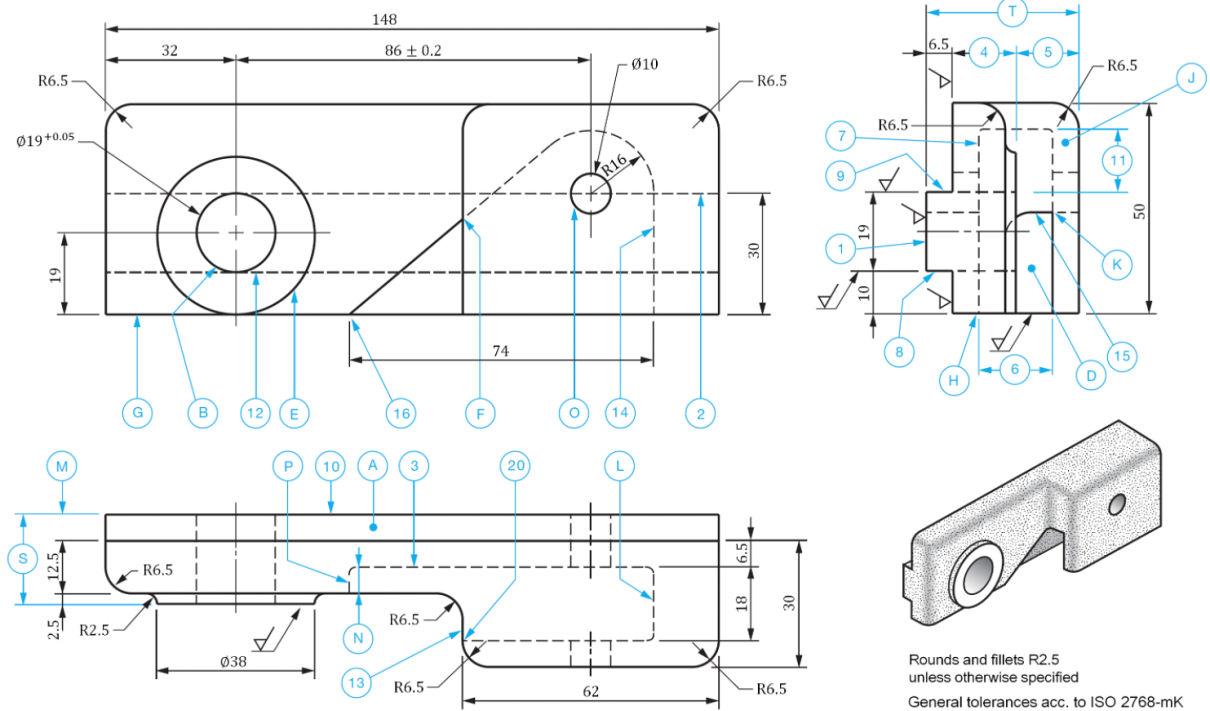


Figure 3.7: Technical drawing

- | | |
|-----------------------------|----------------------------|
| ☐ 9 | ☐ F |
| ☐ 12 | ☐ 2 |

Centre of Knowledge Engineering Design (ED)

Exercise 35:

A tanker truck is half full, see fig. 3.8. The images show three different situations: driving at a constant speed, deceleration, and acceleration. Which assignment of images and situations is correct?



Figure 3.8: Tanker truck

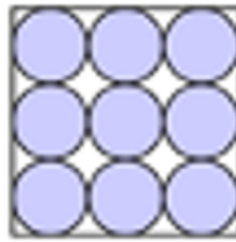
- ☐ Driving at a constant speed (3), deceleration (2), and acceleration (1).



- ☐ Driving at a constant speed (2), deceleration (1), and acceleration (3).
- ☐ Driving at a constant speed (1), deceleration (2), and acceleration (3).
- ☐ Driving at a constant speed (3), deceleration (1), and acceleration (2).

Exercise 36:

In a square with area $A = 1 \text{ m}^2$, n^2 (for $n = 1, 2, 3, \dots$) circles are drawn (see fig. 3.9 for $n = 3$). The area of all summarised circles n is A_n . Which statement is correct?

**Figure 3.9:** Area of circles in a square

- ☐ $A_1 < A_2 < A_4 < A_8$
- ☐ $A_1 > A_2 > A_4 > A_8$
- ☐ $A_1 > A_2 = A_4 > A_8$
- ☐ $A_1 = A_2 = A_4 = A_8$

Exercise 37:

A gear consists of gears A and B (see fig. 3.10). Gear A has Z_A teeth, gear B has Z_B teeth, and gear C has Z_C teeth. While gear Z_A makes n_A revolutions, gear Z_B makes n_B revolutions. Which of the following conclusions is correct?

**Figure 3.10:** Gears

- | | |
|--|--|
| ☐ $n_B = \frac{Z_A}{Z_B \cdot n_A}$ | ☐ $n_B = \frac{Z_A \cdot Z_B}{n_A}$ |
| ☐ $n_B = \frac{Z_B \cdot n_A}{Z_A}$ | ☐ $n_B = \frac{Z_A \cdot n_A}{Z_B}$ |

Exercise 38:

The stiffness ϵ of a coil spring (see fig. 3.11) depends on the core diameter D , the wire diameter d , the number of coils u , and the material property G - the shear modulus. The following applies:

$$\epsilon = \frac{Gd^4}{8uD^5} \quad (3.1)$$

For a second spring made of the same material and with the same number of coils, both the core diameter and the wire diameter are halved. Which statement is correct?

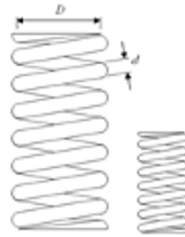


Figure 3.11: Coil spring

- | | |
|--|---|
| ☐ The stiffness ϵ is halved | ☐ The stiffness ϵ is doubled |
| ☐ The stiffness ϵ is unchanged | ☐ The stiffness ϵ is quadrupled |

Exercise 39:

For a production part, a technical sketch shows the top view (DS) and the front view (VA). Which sketch shows the corresponding side view (SA)?

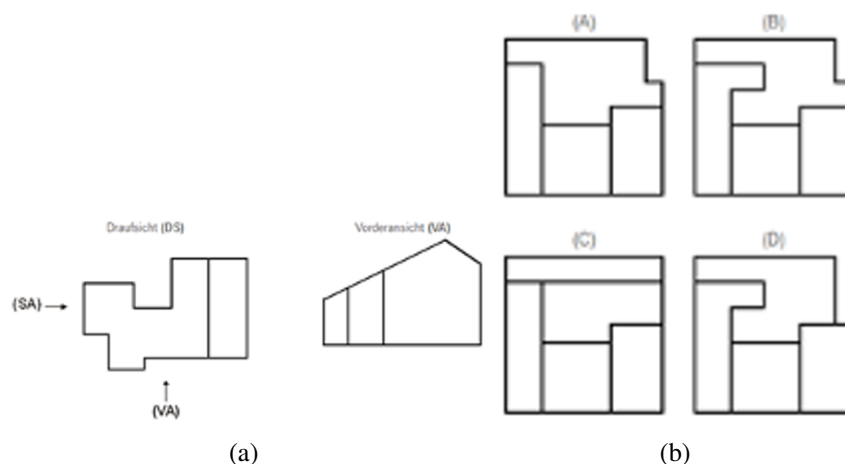


Figure 3.12: Projection views of production part

- | | | | |
|------------------------------|------------------------------|------------------------------|------------------------------|
| ☐ (A) | ☐ (B) | ☐ (C) | ☐ (D) |
|------------------------------|------------------------------|------------------------------|------------------------------|



Table 3.1

Bag #	Number of gummy bears in the bag	Mass of gummy bears in gram
1	5	1.8,2.3,2.0,1.2,2.0
2	6	1.9,2.1,1.9,1.7,2.5,2.0
3	5	1,2.3,2.1,1.95,0.0,2.0
4	6	1.9,1.8,2.1,2.0,2.0,2.1
5	5	2.0,2.1,2.0,2.0,1.9

Centre of Knowledge Fundamental Science and Mathematics (FSM)

Exercise 40:

A factory produces small bags of gummy bears intended as gifts for trade show booths. For technical reasons, it's not always possible to package gummy bears of the same weight. Each bag should contain $(10 \pm 1)\text{g}$ of gummy bears. To check whether the gummy bear production meets the correct specifications, you take five samples, as shown in tab.3.1.

Based on your data analysis, recommend to the production manager which measures should be implemented. Justify your decision(s).

[illegible]

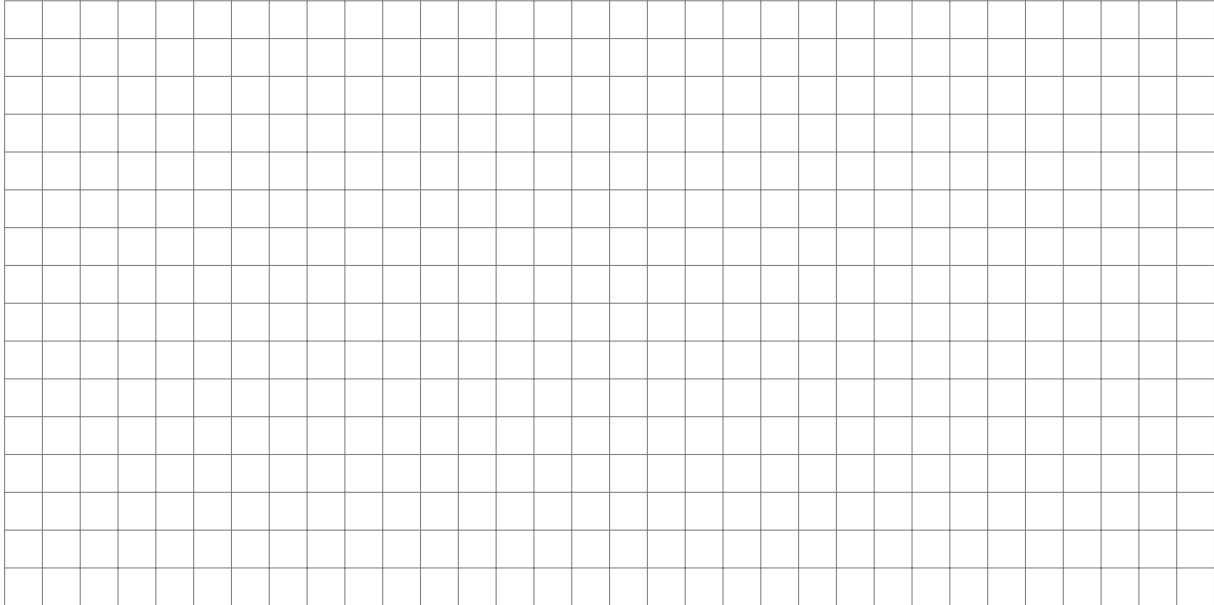
Exercise 41:

A train connection linking Munich airport and Munich city center needs to be optimised. The ride with 11 stops in between takes 41 minutes. Assume that all stops are equally spaced. The train is parking 1 minute at each station. It is driving at a maximum speed of 130 km/h and the maximum acceleration (and deceleration) is 1 m/s². Calculate the distance the train is travelling! Calculate the time the ride would take if there were no stops in between. What is the kinetic friction coefficient that the train wheels experience on the track?

[illegible]

Exercise 42:

Kermit (a frog) is converting energy from his muscles into kinetic energy at an efficiency of 40%. He is hopping under an angle of 30° with respect to the horizontal. By spending 45 mJ he tries to cover a distance of 40 cm and a height of 1 cm. His mass is 50 g. Will he be successful? Sketch the dynamics of the distance $x(t)$, and height $z(t)$ and the trajectory $z(x)$.



Centre of Knowledge Material Strength and Simulation (MSS)

Question loading...



Centre of Knowledge Process Cycles of Energy and Matter (PCEM)

Exercise 43:

Which (smallest natural) number, when divided by 6, gives a remainder of 5, when divided by 5 a remainder of 4, when divided by 4 a remainder of 3 and when divided by 3 a remainder of 2?

Exercise 44:

Systems Thinking: The attached image shows a power plant process for converting heat flows into mechanical power. The lines through the state points 1 to 8 symbolise pipelines through which a constant mass flow of working fluid (water) flows. Which of the following statements are not true for a steady state process?

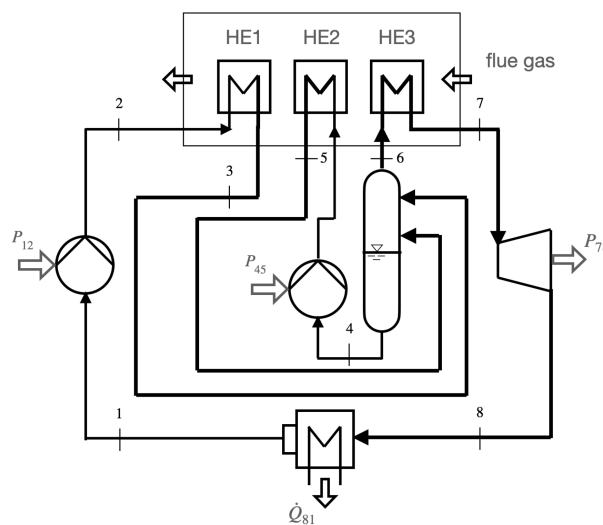


Figure 3.13: Flow scheme of a power plant

- ☐ The mass flow rate through pipe 2 equals the mass flow rate through pipe 6.
- ☐ The mass flow rate through pipe 3 equals the sum of the mass flow rates through pipe 4 and 6.
- ☐ The sum of the mass flows in 4 and 6 is equal to the sum of the mass flows in 5 and 3.
- ☐ The mass flows in 1, 2, 3, 6, 7 and 8 are identical.

Exercise 45:

Engineering approximations: What is the approximate solar radiation power on a soccer pitch on a cloudless summer's day at midday in Ecuador?

- ☐ $P \approx 7 \text{ mW}$
- ☐ $P \approx 7 \text{ kW}$
- ☐ $P \approx 7 \text{ MW}$
- ☐ $P \approx 7 \text{ GW}$

Centre of Knowledge Science Communication (SC)

Question loading...

Results

Rate your personal performance: _____%



Improving Assessment Questions on Skills Reflecting on the questions asked, considering their purpose on evaluating the needed competences for the study course EfS: What is your opinion on these and what should be done for improvement? What to keep/drop/try?

1. General opinion on feasibility:

[illegible]

- ## 2. Things worth keeping:

[illegible]

- ### 3. Content to drop:

[illegible]

4. Next time, try to...

[illegible]

5. A question to add:

[illegible]

Thank you very much for your feedback

1.3 Competences of Engineers for Sustainability

The sustainable transformation requires a special set of engineering skills that go beyond the technical aspects covered in previous requirements. There is no simple answer to complex questions. Sustainability is not just a technical problem; it also involves decision-making that considers political, ethical and normative aspects, see tab.3.2

Table 3.2: Sustainable development - Facts and standards, taken from [39], translated to english

	Scientific, Technical	Normative, Ethical, Political, Goal-oriented
	Facts	Values
Past Analytical	How do technical developments work? (laws of nature, physics) →	What should the world be like? (ethics, meta-morale) ↓
Future Strategic	How can we shape the future? (processes, interrelationships) ↓	What future do we want? (morale, regulations, laws) ←
Action Practical	Implementation (facts, figures, actions) →	What do I want? What can I do? (evaluation, decision)

To tackle theses complex challenges, in [57], three key competencies for transformation are named:

- cognitive understanding of complex problem contexts
- emotional intelligence
- and the ability to convert intentions into action



SDGs and Inner Development Goals

For a successful implementation of the SDGs also an inner setting has to be crafted [19]. Or, as at times stated 'the outer transformation is connected to an inner transformation'. Or as quoted in [3]:

'The Inner Development Goals Framework simplifies a complex field of human development to help us better identify understand communicate develop integrate the inner skills needed for sustainable development.'

An overview is given in fig. 3.14



Figure 3.14: Inner Development Goals Framework, taken from [3]

Exercise 46:

Explore the IDGs under <https://innerdevelopmentgoals.org/framework/#explore>. How do rate your own path reaching these inner development goals? If help is needed for your self assessment, use <https://idg.gemeinwohltun.ch> Afterwards, make a short statement on the relevance of the IDGs (your opinion!).

1.4 Preparing for the field trip - General Overview on the Company

Check the homepage of the company at hand for general information. Also take other sources of informations into account (provided material, www, library). Use the following questions for orientation, add questions as desired.

Exercise 47:

1. In which business sector is the company located? What is its size (SMU or large company) and what is the companies legal form (AG, KGaA, SE , GmbH)?
2. Which are the direct competing companies? Where are these located?
3. Which products are provided for which customers? What are the special requirements of the customers and how is that addressed in product development?
4. What resources are mostly used for the production of the products? How are the products manufactured? Which machines and which personnel is needed in the company?

1.5 Preparing for the field trip - What skills are needed most?

Considering your knowledge on skills and competences of an engineer, analyse the information of the company as a potential employer for you as a graduate. Some topics to clarify:

Exercise 48:

1. What 'typical' engineering specialisations would you expect to encounter in the company? List specialisations and rate by '+o-'
2. Which software-tools are supposedly used in the engineering department (AutoCAD, EPlan, Fluent, SAP, ERP, Gaussian etc.)?
3. What other vocations do you expect to have interfaces with, sort them according to direct and indirect interaction. Also consider the information on vocational training programmes provided (what is their respective responsibility?).
4. What typical incomes would you expect in the company as an entry level engineer?
5. What typical engineering courses are most relevant for working in the company? Which are not relevant at all (check module handbooks of classical engineering courses)?
6. What competences not present in a typical engineering study program curriculum are additionally needed for a fulfilling work position in the company?
7. Check the job offerings page? How many engineers for sustainability are wanted at the moment? How would you apply for a position (prepare key-points for a motivational letter for a potential application)?



Exercise 49:

When specifically aiming for a job at the given company; what would your study plan in EfS look like? Use the template in fig. 3.15 and enter the centres abbreviation. Give reason for the choices you make; also consider the information provided in the module handbook EfS.

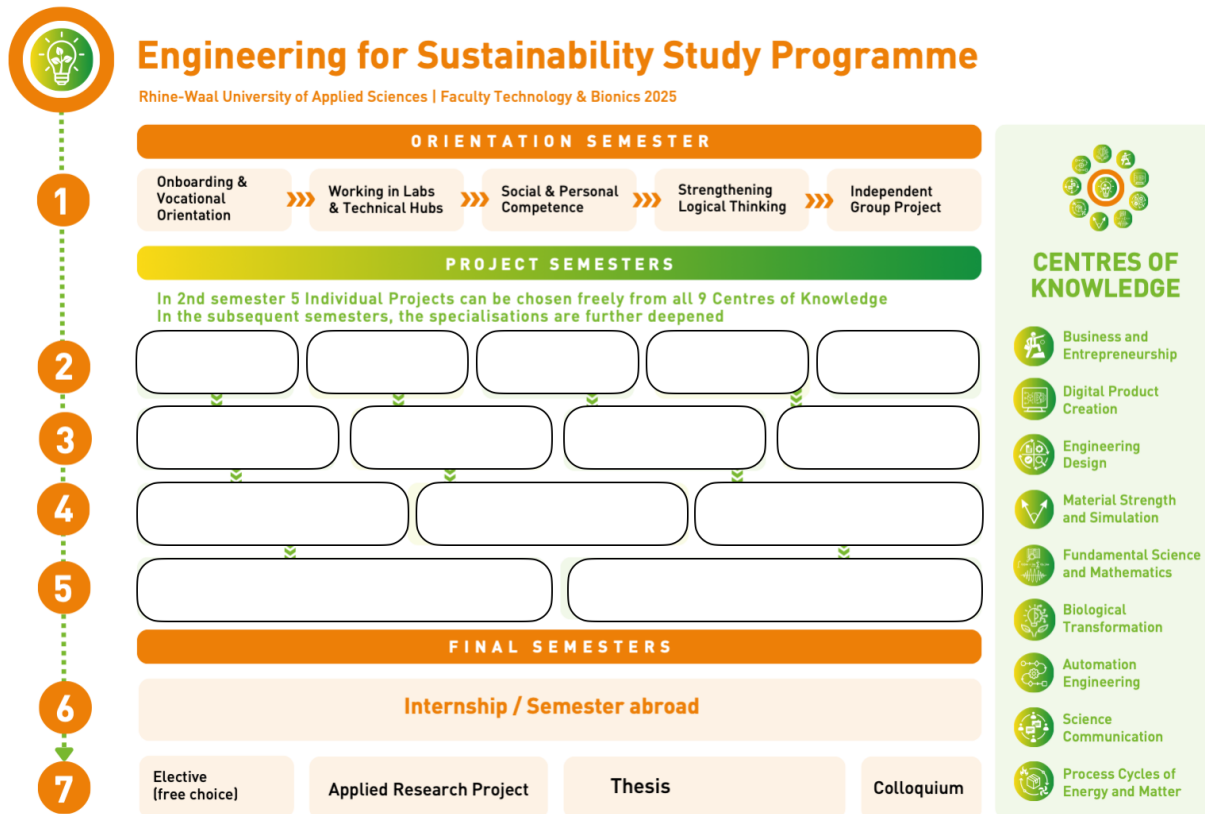


Figure 3.15: Study plan, taken from [26], modified (added white space for entering center abbreviations)

1.6 Where have the lectures gone? What is this 'Center of Knowledge' ?

Here one of the major aspects of the study course shall be addressed, the aim is to build up competences. This is done by projects instead of classical lectures. In the latter, usually separate topics are taught (from the beginning to the end) within equally separate courses, namely *thermodynamics, physics, mechanics, programming, electronics etc.*. In a typical engineering project however, the responsible person has to combine all skills at hand (some more some less) to get the project done - how that is done is left to the project leader and usually not addressed in the classical study courses (as also the teaching personell is highly specialised and not necessarily capable of properly teaching at interfaces to other disciplines).

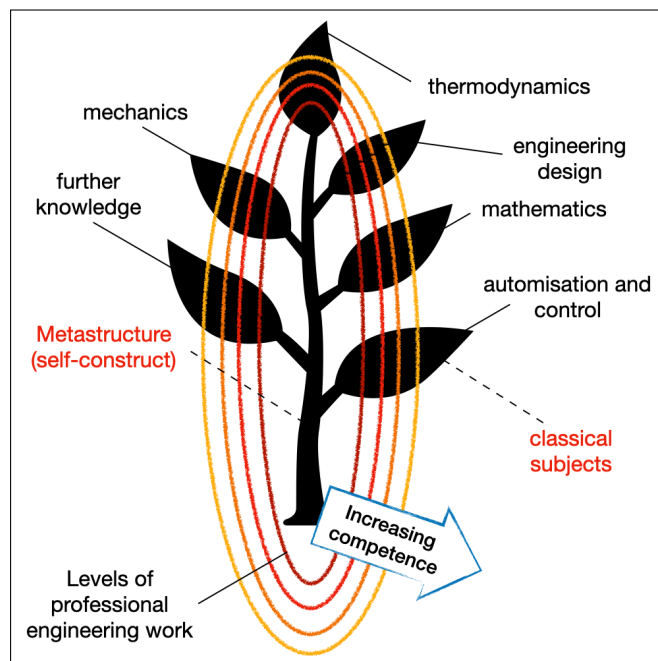


Figure 3.16: Competences and knowledge

As depicted in fig. 3.16, within a given project, one needs knowledge from different subjects, the more complicated the project is, the more knowledge is needed and the more competences are required to combine these for a good performance. These 'levels' shown are trained within the respective semester of the given study course, aiming at a high level or professional competence directly available to the graduates. The depiction in fig. 3.16 is oversimplifying reality in one central aspect, the 'leaves' at the tree are not singular subjects but would be more accurately represented by an set of skills, e.g. a skillset called 'mechanics' consists of sub skills like 'dynamics', 'statics' and 'material strength'. In the end this is leading to a fractal geometry instead of the shown leaves. within our 'centers of knowledge' we have grouped sets of skills usually required in a given engineering sub-discipline, teaching the sub-skills (former 'courses') all at once, but only as much as it is needed at the moment. E.g. instead of fully going through the courses thermodynamics, fluid mechanics, process control, etc. sequentially, fractions of the content are taken and addressed as needed for the given project. An enhanced understanding of the respective subjects is build up in the next (more complicated) project of the next semester and so forth. In the end, the same content is covered as in 'classical' study courses - only in a different temporal structure. Which competences of which discipline to enhance is your choice, guided by your personal mentor (see the respective chapter on further information on the mentoring process).



Relations between Centres of Knowledge and classical engineering skills

'Classical' engineering skills are not obsolete when addressing future skills for engineers, on the contrary these remain the basis for profound and professional engineering work. As stated above, the *centres of knowledge* are grouping content of other classical modules into one organisational unit, see fig. 3.17

9 CENTRES OF KNOWLEDGE



Figure 3.17: Centres of Knowledge

In the following section, organise the information that you have (or can get) on each center, also make a preliminary rating on how the goals, topics and content of each center applies to your personal preferences (after gathering information ;-)).



SUSTAINABLE PRODUCT DEVELOPMENT BETWEEN TECHNOLOGY, CREATIVITY, AND PRACTICE

How are sustainable products created? In this Centre, students learn to design technical components and systems so that they are functional, resource-efficient, and future-proof – from material selection to “second life.” We work interdisciplinarily and practice-oriented on solutions for a circular industry.

- Project work with a focus on sustainable innovation
- Design methods & creativity techniques
- Material selection based on sustainability criteria
- Second-life strategies and circular economy
- Evaluation of products according to resource efficiency
- Fundamentals of technical project management



Figure 3.18: Centres of Knowledge: Engineering Design

Exercise 50:

Check the module handbook and other sources. What are (in your opinion) the most important competencies that can be acquired in the center?

Exercise 51:

Which content of 'typical' engineering courses / modules would you expect within this center?

Exercise 52:

How relevant is the content of the center for sustainable engineering? Give reason .

Exercise 53:

Try to imagine a sustainability focussed project that fits into the center (title only).

Exercise 54:

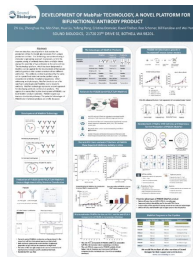
Wow! I can totally see me working in that field! (Why?)

(nah... not really)	1	2	3	4	5	6	7	8	9	10	(yea! totally!)
---------------------	---	---	---	---	---	---	---	---	---	----	-----------------



SCIENCE COMMUNICATION FOR A SUSTAINABLE SOCIETY

In this Centre, students learn to critically analyse scientific information, prepare it in an understandable way, and communicate it to different audiences across various media. Ethical standards, creative methods, and social impact are at the core.



- Assessing the quality of scientific information
- Developing targeted communication strategies
- Designing media content for different audiences
- Data visualisation & data sonification
- Media art & participatory formats for public engagement

Figure 3.19: Centres of Knowledge: Science Communication

Exercise 55:

Check the module handbook and other sources. What are (in your opinion) the most important competencies that can be acquired in the center?

Exercise 56:

Which content of 'typical' engineering courses / modules would you expect within this center?

Exercise 57:

How relevant is the content of the center for sustainable engineering? Give reason .

Exercise 58:

Try to imagine a sustainability focussed project that fits into the center (title only).

Exercise 59:

Wow! I can totally see me working in that field! (Why?)

(nah... not really)	1	2	3	4	5	6	7	8	9	10	(yea! totally!)
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ENGINEERING FOR SUSTAINABILITY

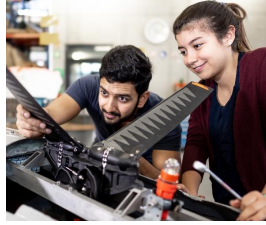


Biological Transformation



LEARNING FROM NATURE – RETHINKING TECHNOLOGY SUSTAINABLY

In the “Biological Transformation” Centre, we develop new technical concepts inspired by biological principles. From natural material cycles to robotics, students learn how ecology and technology can be considered together.



- Understanding sustainability in nature – with microscopy & field research
- Sensors & actuators inspired by nature (e.g., artificial eyes & ears)
- Analysing biomechanics & form-function principles in animals
- Robotics: swimming robots, flight systems, animal-inspired drones
- Applications in rehabilitation technology or sports innovation (e.g., exoskeletons)

Figure 3.20: Centres of Knowledge: Biological Transformation

Exercise 60:

Check the module handbook and other sources. What are (in your opinion) the most important competencies that can be acquired in the center?

Exercise 61:

Which content of 'typical' engineering courses / modules would you expect within this center?

Exercise 62:

How relevant is the content of the center for sustainable engineering? Give reason .

Exercise 63:

Try to imagine a sustainability focussed project that fits into the center (title only).

Exercise 64:

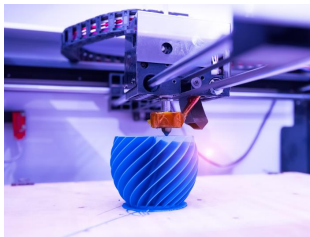
Wow! I can totally see me working in that field! (Why?)

(nah... not really)	1	2	3	4	5	6	7	8	9	10	(yea! totally!)
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SUSTAINABLE PRODUCT DEVELOPMENT WITH DIGITAL TOOLS

In the Centre of Digital Product Creation, we develop mechanical products from idea to virtual prototype – digital, efficient, and resource-saving. We work with modern 3D technologies and learn how product development processes can be automated, optimised, and designed sustainably.



- End-to-end product development with digital tools
- 3D CAD modelling, 3D scanning & 3D printing
- Automation of design processes
- Development of sustainable product concepts
- Digital teamwork & data management
- Projects focusing on efficiency & innovation

Figure 3.21: Centres of Knowledge: Digital Product Creation

Exercise 65:

Check the module handbook and other sources. What are (in your opinion) the most important competencies that can be acquired in the center?

Exercise 66:

Which content of 'typical' engineering courses / modules would you expect within this center?

Exercise 67:

How relevant is the content of the center for sustainable engineering? Give reason .

Exercise 68:

Try to imagine a sustainability focussed project that fits into the center (title only).

Exercise 69:

Wow! I can totally see me working in that field! (Why?)

(nah... not really)	1	2	3	4	5	6	7	8	9	10	(yea! totally!)
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ENGINEERING FOR SUSTAINABILITY



Fundamental Science & Mathematics



**UNDERSTANDING, APPLYING, AND CONNECTING
MATHEMATICAL AND PHYSICAL FUNDAMENTALS**

In this Centre, students learn to describe technical and natural systems precisely – with mathematics, physics, programming, and scientific methodology. We work on projects that combine measurement, simulation, analysis, and documentation.

- Mathematical modelling & physical system description
- Statistics, data analysis & numerical methods
- Programming fundamentals for technical applications
- Scientific writing & working methods
- Project example: Building a Raman spectrometer with practical applications

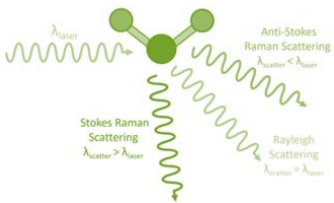


Figure 3.22: Centres of Knowledge: Fundamental Science and Mathematics

Exercise 70:

Check the module handbook and other sources. What are (in your opinion) the most important competencies that can be acquired in the center?

Exercise 71:

Which content of 'typical' engineering courses / modules would you expect within this center?

Exercise 72:

How relevant is the content of the center for sustainable engineering? Give reason .

Exercise 73:

Try to imagine a sustainability focussed project that fits into the center (title only).

Exercise 74:

Wow! I can totally see me working in that field! (Why?)

(nah... not really)	1	2	3	4	5	6	7	8	9	10	(yea! totally!)
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AUTOMATING, CONTROLLING, AND OPTIMISING SYSTEMS

In the "Automation Engineering" Centre of Knowledge, students learn how technical systems can function autonomously through sensors, actuators, and intelligent control. We develop solutions to automate machines efficiently, safely, and user-friendly.

- Measuring, controlling, and regulating technical systems
- Simulation & modelling with Matlab/Simulink
- Use of PLCs, Arduino & others for process control
- Optimisation of energy consumption & system efficiency
- Design of human-machine interfaces
- Safety aspects of automated systems



Figure 3.23: Centres of Knowledge: Automation Engineering

Exercise 75:

Check the module handbook and other sources. What are (in your opinion) the most important competencies that can be acquired in the center?

Exercise 76:

Which content of 'typical' engineering courses / modules would you expect within this center?

Exercise 77:

How relevant is the content of the center for sustainable engineering? Give reason .

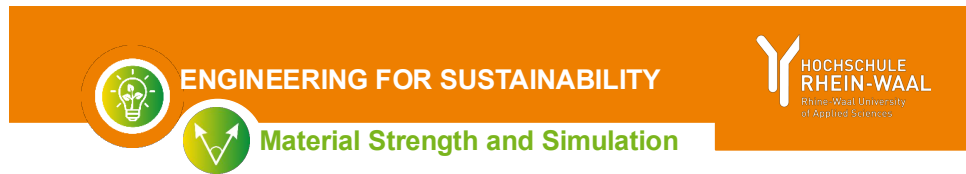
Exercise 78:

Try to imagine a sustainability focussed project that fits into the center (title only).

Exercise 79:

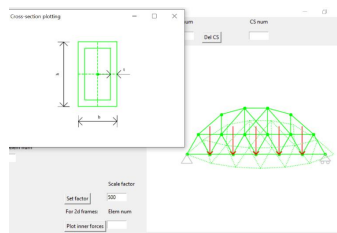
Wow! I can totally see me working in that field! (Why?)

(nah... not really)	1	2	3	4	5	6	7	8	9	10	(yea! totally!)
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HOW FORCES ACT – AND WHAT THEY DO TO STRUCTURES

Here, students learn how technical structures respond under load – whether they deform or move. Using methods of engineering mechanics, we analyse and simulate whether constructions deliver what they promise.



- Understanding the basics of statics & dynamics
- Structural analysis with paper, pencil – and Python
- Testing the load-bearing capacity of technical components
- Simulating deformation & motion
- Developing safe & efficient constructions

Figure 3.24: Centres of Knowledge: Material Strength and Simulation

Exercise 80:

Check the module handbook and other sources. What are (in your opinion) the most important competencies that can be acquired in the center?

Exercise 81:

Which content of 'typical' engineering courses / modules would you expect within this center?

Exercise 82:

How relevant is the content of the center for sustainable engineering? Give reason .

Exercise 83:

Try to imagine a sustainability focussed project that fits into the center (title only).

Exercise 84:

Wow! I can totally see me working in that field! (Why?)

(nah... not really)	1	2	3	4	5	6	7	8	9	10	(yea! totally!)
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SUSTAINABLE BUSINESS, ENTREPRENEURIAL ACTION, SHAPING THE FUTURE

Here, students learn how economic activity can be combined with ecological responsibility. In practice-oriented projects, we develop sustainable business models, analyse business processes, and design products along the entire life cycle.

- Introduction to sustainability & systems thinking
- Circular economy & resource management
- Sustainable business models & strategies
- Full product life cycle management
- Innovation management & sustainable entrepreneurship



Figure 3.25: Centres of Knowledge: Business and Entrepreneurship

Exercise 85:

Check the module handbook and other sources. What are (in your opinion) the most important competencies that can be acquired in the center?

Exercise 86:

Which content of 'typical' engineering courses / modules would you expect within this center?

Exercise 87:

How relevant is the content of the center for sustainable engineering? Give reason .

Exercise 88:

Try to imagine a sustainability focussed project that fits into the center (title only).

Exercise 89:

Wow! I can totally see me working in that field! (Why?)

(nah... not really)	1	2	3	4	5	6	7	8	9	10	(yea! totally!)
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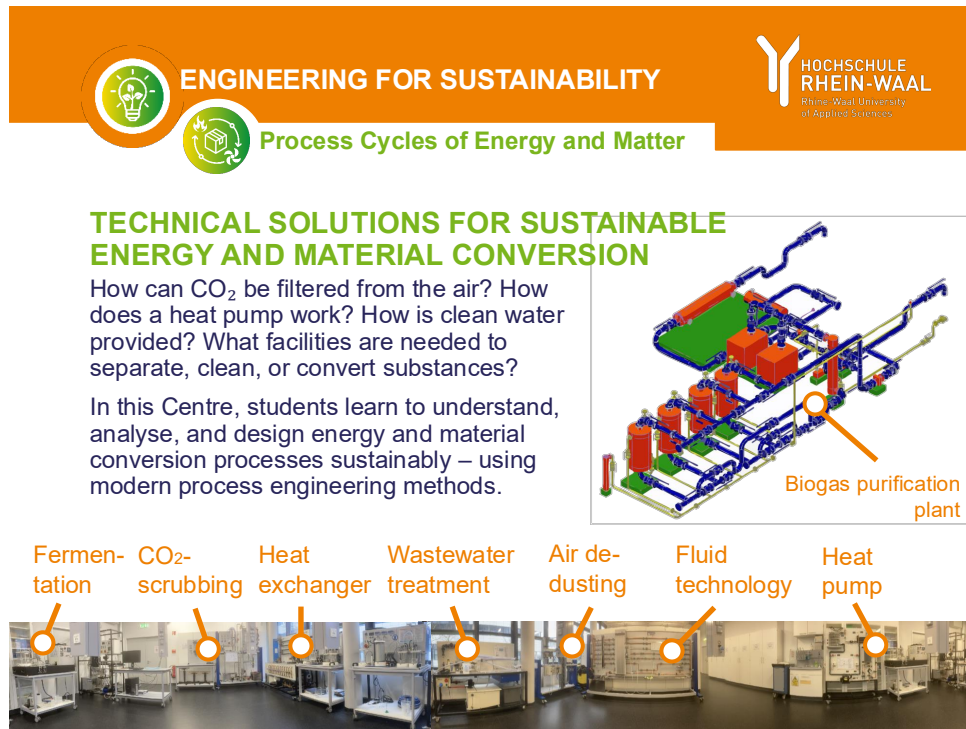


Figure 3.26: Centres of Knowledge: Process Cycles of Energy and Matter

Exercise 90:

Check the module handbook and other sources. What are (in your opinion) the most important competencies that can be acquired in the center?

Exercise 91:

Which content of 'typical' engineering courses / modules would you expect within this center?

Exercise 92:

How relevant is the content of the center for sustainable engineering? Give reason .

Exercise 93:

Try to imagine a sustainability focussed project that fits into the center (title only).

Exercise 94:

Wow! I can totally see me working in that field! (Why?)

(nah... not really)	1	2	3	4	5	6	7	8	9	10	(yea! totally!)
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Exercise 95:

In the end, in a sustainability project, content belonging to all the mentioned centres are usually needed. Can you imagine a sustainability engineering project, incorporating all of the centres with their characteristic skills and competences? Sketch the 'perfect' project:

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[7] [30] [8] [9] [31] [43] [19] [32] [34] [42] [44] [48] [57] [59] [61] [62] [63] [64] [67] [68] [69] [65] [71]